

§3. PRODUCTS OF PRESCHEMES

3.1. Sums of preschemes Let (X_α) be any family of preschemes; let X be a topological space which is the *sum* of the underlying spaces X_α ; X is then the union of pairwise disjoint open subspaces X'_α , and for each α there is a homeomorphism ϕ_α from X_α to X'_α . If we equip each of the X'_α with the sheaf $(\phi_\alpha)_*(\mathcal{O}_{X_\alpha})$, it is clear that X becomes a prescheme, which we call the *sum* of the family of preschemes (X_α) and which we denote $\bigsqcup_\alpha X_\alpha$. If Y is a prescheme, then the map $f \mapsto (f \circ \phi_\alpha)$ is a *bijection* from the set $\text{Hom}(X, Y)$ to the product set $\prod_\alpha \text{Hom}(X_\alpha, Y)$. In particular, if the X_α are S -preschemes, with structure morphisms ψ_α , then X is an S -prescheme by the unique morphism $\psi : X \rightarrow S$ such that $\psi \circ \phi_\alpha = \psi_\alpha$ for each α . The sum of two preschemes X and Y is denoted by $X \sqcup Y$. It is immediate that, if $X = \text{Spec}(A)$ and $Y = \text{Spec}(B)$, then $X \sqcup Y$ is canonically identified with $\text{Spec}(A \times B)$.

3.2. Products of preschemes

Definition (3.2.1). — Given S -preschemes X and Y , we say that a triple (Z, p_1, p_2) , consisting of an S -prescheme Z , and S -morphisms $p_1 : Z \rightarrow X$ and $p_2 : Z \rightarrow Y$, is a *product* of the S -preschemes X and Y , if, for each S -prescheme T , the map $f \mapsto (p_1 \circ f, p_2 \circ f)$ is a bijection from the set of S -morphisms from T to Z , to the set of pairs consisting of an S -morphism $T \rightarrow X$ and an S -morphism $T \rightarrow Y$ (in other words, a bijection

I | 105

$$\text{Hom}_S(T, Z) \simeq \text{Hom}_S(T, X) \times \text{Hom}_S(T, Y).$$

This is the general notion of a *product* of two objects in a category, applied to the category of S -preschemes (T, I, 1.1); in particular, a product of two S -preschemes is *unique* up to a unique S -isomorphism. Because of this uniqueness, most of the time we will denote a product of two S -preschemes X and Y by $X \times_S Y$ (or simply $X \times Y$, when there is no chance of confusion), with the morphisms p_1 and p_2 (the *canonical projections* of $X \times_S Y$ to X and to Y , respectively) being suppressed in the notation. If $g : T \rightarrow X$ and $h : T \rightarrow Y$ are S -morphisms, we denote by $(g, h)_S$, or simply (g, h) , the S -morphism $f : T \rightarrow X \times_S Y$ such that $p_1 \circ f = g$, $p_2 \circ f = h$. If X' and Y' are two S -preschemes, p'_1 and p'_2 the canonical projections of $X' \times_S Y'$ (assumed to exist), and $u : X' \rightarrow X$ and $v : Y' \rightarrow Y$ are S -morphisms, we write $u \times_S v$ (or simply $u \times v$) for the S -morphism $(u \circ p'_1, v \circ p'_2)_S$ from $X' \times_S Y'$ to $X \times_S Y$.

When S is an affine scheme given by some ring A , we often replace S by A in the above notation.

Proposition (3.2.2). — Let X, Y , and S be affine schemes, given by rings B, C , and A (respectively). Let $Z = \text{Spec}(B \otimes_A C)$, and let p_1 and p_2 be the S -morphisms corresponding (2.2.4) to the canonical A -homomorphisms $u : b \mapsto b \otimes 1$ and $v : c \mapsto 1 \otimes c$ (respectively) from B and C to $B \otimes_A C$; then (Z, p_1, p_2) is a product of X and Y .

Proof. According to (2.2.4), it suffices to check that, if, to each A -homomorphism $f : B \otimes_A C \rightarrow L$ (where L is an A -algebra), we associate the pair $(f \circ u, f \circ v)$, then this defines a bijection $\text{Hom}_A(B \otimes_A C, L) \simeq \text{Hom}_A(B, L) \times \text{Hom}_A(C, L)$,¹ which follows immediately from the definitions and the fact that $b \otimes c = (b \otimes 1)(1 \otimes c)$. $\square$

Corollary (3.2.3). — Let T be an affine scheme given by some ring D , and $\alpha = ({}^a\rho, \tilde{\rho})$ (resp. $\beta = ({}^a\sigma, \tilde{\sigma})$) an S -morphism $T \rightarrow X$ (resp. $T \rightarrow Y$), where ρ (resp. σ) is an A -homomorphism from B (resp. C) to D ; then $(\alpha, \beta)_S = ({}^a\tau, \tilde{\tau})$, where τ is the homomorphism $B \otimes_A C \rightarrow D$ such that $\tau(b \otimes c) = \rho(b)\sigma(c)$.

Proposition (3.2.4). — Let $f : S' \rightarrow S$ be a monomorphism of preschemes (T, I, 1.1), and let X and Y be S' -preschemes, also considered as S -preschemes via f . Every product of the S -preschemes X and Y is then a product of the S' -preschemes X and Y , and vice versa.

Proof. Let $\phi : X \rightarrow S'$ and $\psi : Y \rightarrow S'$ be the structure morphisms. If T is an S -prescheme, and $u : T \rightarrow X$ and $v : T \rightarrow Y$ are S -morphisms, then we have, by definition, that $f \circ \phi \circ u = f \circ \psi \circ v = \theta$, the structure morphism of T ; the hypothesis on f implies that $\phi \circ u = \psi \circ v = \theta'$, and so we can consider T as an S' -prescheme with structure morphism θ' , and u and v as S' -morphisms. The conclusion of the proposition follows immediately, taking (3.2.1) into account. $\square$

¹The notation Hom_A denotes here the set of homomorphisms of A -algebras.

Corollary (3.2.5). — Let X and Y be S -preschemes, with structure morphisms $\phi : X \rightarrow S$ and $\psi : Y \rightarrow S$, and let S' be an open subset of S such that $\phi(X) \subset S'$ and $\psi(Y) \subset S'$. Every product of the S -preschemes X and Y is then also a product of the S' -preschemes X and Y , and conversely.

It suffices to apply (3.2.4) to the canonical injection $S' \rightarrow S$.

I | 106

Theorem (3.2.6). — Given S -preschemes X and Y , there exists a product $X \times_S Y$.

The proof proceeds in several steps.

Lemma (3.2.6.1). — Let (Z, p, q) be a product of X and Y , and U and V open subsets of X and Y , respectively. If we let $W = p^{-1}(U) \cap q^{-1}(V)$, then the triple consisting of W and the restrictions of p and q to W (considered as the morphisms $W \rightarrow U$ and $W \rightarrow V$, respectively) is a product of U and V .

Indeed, if T is an S -prescheme, then we can identify the S -morphisms $T \rightarrow W$ with the S -morphisms $T \rightarrow Z$ mapping T to W . Then, if $g : T \rightarrow U$ and $h : T \rightarrow V$ are any two S -morphisms, we can consider them as S -morphisms from T to X and to Y , respectively, and, by hypothesis, there is then a unique S -morphism $f : T \rightarrow Z$ such that $g = p \circ f$ and $h = q \circ f$. Since $p(f(T)) \subset U$ and $q(f(T)) \subset V$, we have

$$f(T) \subset p^{-1}(U) \cap q^{-1}(V) = W,$$

hence our claim.

Lemma (3.2.6.2). — Let Z be an S -prescheme, $p : Z \rightarrow X$ and $q : Z \rightarrow Y$ both S -morphisms, (U_α) an open cover of X , and (V_λ) an open cover of Y . Suppose that, for each pair (α, λ) , the S -prescheme $W_{\alpha\lambda} = p^{-1}(U_\alpha) \cap q^{-1}(V_\lambda)$ and the restrictions of p and q to $W_{\alpha\lambda}$ form a product of U_α and V_λ . Then (Z, p, q) is a product of X and Y .

We first show that, if f_1 and f_2 are S -morphisms $T \rightarrow Z$, then the equations $p \circ f_1 = p \circ f_2$ and $q \circ f_1 = q \circ f_2$ imply that $f_1 = f_2$. Indeed, Z is the union of the $W_{\alpha\lambda}$, so the $f_1^{-1}(W_{\alpha\lambda})$ form an open cover of T , and similarly for $f_2^{-1}(W_{\alpha\lambda})$. In addition, we have

$$f_1^{-1}(W_{\alpha\lambda}) = f_1^{-1}(p^{-1}(U_\alpha) \cap q^{-1}(V_\lambda)) = f_2^{-1}(p^{-1}(U_\alpha) \cap q^{-1}(V_\lambda)) = f_2^{-1}(W_{\alpha\lambda})$$

by hypothesis, and it thus reduces to noting that the restrictions of f_1 and f_2 to $f_1^{-1}(W_{\alpha\lambda}) = f_2^{-1}(W_{\alpha\lambda})$ are identical for each pair of indices. But since these restrictions can be considered as S -morphisms from $f_1^{-1}(W_{\alpha\lambda})$ to $W_{\alpha\lambda}$, our claim follows from the hypotheses and Definition (3.2.1).

Suppose now that we are given S -morphisms $g : T \rightarrow X$ and $h : T \rightarrow Y$. Let $T_{\alpha\lambda} = g^{-1}(U_\alpha) \cap h^{-1}(V_\lambda)$; then the $T_{\alpha\lambda}$ form an open cover of T . By hypothesis, there exists an S -morphism $f_{\alpha\lambda}$ such that $p \circ f_{\alpha\lambda}$ and $q \circ f_{\alpha\lambda}$ are, respectively, the restrictions of g and h to $T_{\alpha\lambda}$. We will now show that the restrictions of $f_{\alpha\lambda}$ and $f_{\beta\mu}$ to $T_{\alpha\lambda} \cap T_{\beta\mu}$ coincide, which will finish the proof of Lemma (3.2.6.2). The images of $T_{\alpha\lambda} \cap T_{\beta\mu}$ under $f_{\alpha\lambda}$ and $f_{\beta\mu}$ are contained in $W_{\alpha\lambda} \cap W_{\beta\mu}$ by definition. Since

$$W_{\alpha\lambda} \cap W_{\beta\mu} = p^{-1}(U_\alpha \cap U_\beta) \cap q^{-1}(V_\lambda \cap V_\mu),$$

it follows from Lemma (3.2.6.1) that $W_{\alpha\lambda} \cap W_{\beta\mu}$ and the restrictions to this prescheme of p and q form a product of $U_\alpha \cap U_\beta$ and $V_\lambda \cap V_\mu$. Since $p \circ f_{\alpha\lambda}$ and $p \circ f_{\beta\mu}$ coincide on $T_{\alpha\lambda} \cap T_{\beta\mu}$ and similarly for $q \circ f_{\alpha\lambda}$ and $q \circ f_{\beta\mu}$, we see that $f_{\alpha\lambda}$ and $f_{\beta\mu}$ coincide on $T_{\alpha\lambda} \cap T_{\beta\mu}$. $\square$

Lemma (3.2.6.3). — Let (U_α) be an open cover of X , (V_λ) an open cover of Y , and suppose that, for each pair (α, λ) , there exists a product of U_α and V_λ ; then there exists a product of X and Y .

I | 107

Applying Lemma (3.2.6.1) to the open sets $U_\alpha \cap U_\beta$ and $V_\lambda \cap V_\mu$, we see that there exists a product of S -preschemes induced, respectively, by X and Y on these open sets; in addition, the uniqueness of the product shows that, if we set $i = (\alpha, \lambda)$ and $j = (\beta, \mu)$, then there is a canonical isomorphism h_{ij} (resp. h_{ji}) from this product to an S -prescheme W_{ij} (resp. W_{ji}) induced by $U_\alpha \times_S V_\lambda$ (resp. $U_\beta \times_S V_\mu$) on an open set; then $f_{ij} = h_{ij} \circ h_{ji}^{-1}$ is an isomorphism from W_{ji} to W_{ij} . In addition, for a third pair $k = (\gamma, \nu)$, we have $f_{ik} = f_{ij} \circ f_{jk}$ on $W_{ki} \cap W_{kj}$, by applying Lemma (3.2.6.1) to the open sets $U_\alpha \cap U_\beta \cap U_\gamma$ and $V_\lambda \cap V_\mu \cap V_\nu$ in U_β and V_μ , respectively. It follows that we have a prescheme Z , an open cover (Z_i) of the underlying space of Z , and, for each i , an isomorphism g_i from the induced prescheme Z_i to the prescheme $U_\alpha \times_S V_\lambda$, so that, for each pair (i, j) , we have

$f_{ij} = g_i \circ g_j^{-1}$ (2.3.1); in addition, we have $g_i(Z_i \cap Z_j) = W_{ij}$. Let p_i and q_i be the projections, and θ_i the structure morphism, of the S -prescheme $U_\alpha \times_S V_\lambda$. We immediately see that $p_i \circ g_i = p_j \circ g_j$ on $Z_i \cap Z_j$, and similarly for the two other morphisms. We can thus define the morphisms of preschemes $p : Z \rightarrow X$ (resp. $q : Z \rightarrow Y$, $\theta : Z \rightarrow S$) by the condition that p (resp. q , θ) coincide with $p_i \circ g_i$ (resp. $q_i \circ g_i$, $\theta_i \circ g_i$) on each of the Z_i ; Z , equipped with θ , is then an S -prescheme. We now show that $Z'_i = p^{-1}(U_\alpha) \cap q^{-1}(V_\lambda)$ is equal to Z_i . For each index $j = (\beta, \mu)$, we have $Z_j \cap Z'_i = g_j^{-1}(p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda))$. We have

$$p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda) = p_j^{-1}(U_\alpha \cap U_\beta) \cap q_j^{-1}(V_\lambda \cap V_\mu);$$

By Lemma (3.2.6.1), the restrictions of p_j and q_j to $p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda)$ define, on this S -prescheme, the structure of a product of $U_\alpha \cap U_\beta$ and $V_\lambda \cap V_\mu$; but the uniqueness of the product then implies that $p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda) = W_{ji}$. As a result, we have $Z_j \cap Z'_i = Z_j \cap Z_i$ for each j , hence $Z'_i = Z_i$. We then deduce from Lemma (3.2.6.2) that (Z, p, q) is a product of X and Y .

Lemma (3.2.6.4). — *Let $\phi : X \rightarrow S$ and $\psi : Y \rightarrow S$ be the structure morphisms of X and Y , (S_i) an open cover of S , and let $X_i = \phi^{-1}(S_i)$, $Y_i = \psi^{-1}(S_i)$. If each of the products $X_i \times_S Y_i$ exists, then $X \times_S Y$ exists.*

By Lemma (3.2.6.3), it suffices to prove that the products $X_i \times_S Y_j$ exist for all i and j . Set $X_{ij} = X_i \cap X_j = \phi^{-1}(S_i \cap S_j)$, $Y_{ij} = Y_i \cap Y_j = \psi^{-1}(S_i \cap S_j)$; by Lemma (3.2.6.1), the product $Z_{ij} = X_{ij} \times_S Y_{ij}$ exists. We now note that, if T is an S -prescheme, and if $g : T \rightarrow X_i$ and $h : T \rightarrow Y_j$ are S -morphisms, then we necessarily have that $\phi(g(T)) = \psi(h(T)) \subset S_i \cap S_j$ by the definition of an S -morphism, and thus that $g(T) \subset X_{ij}$ and $h(T) \subset Y_{ij}$; it is then immediate that Z_{ij} is the product of X_i and Y_j .

(3.2.6.5). We can now complete the proof of Theorem (3.2.6). If S is an affine scheme, then there are covers (U_α) and (V_λ) of X and Y (respectively) consisting of affine open subsets; since $U_\alpha \times_S V_\lambda$ exists by (3.2.2), Lemma (3.2.6.3) shows that $X \times_S Y$ exists. If S is any prescheme, then there is a cover (S_i) of S consisting of affine open subsets. If $\phi : X \rightarrow S$ and $\psi : Y \rightarrow S$ are the structure morphisms, and if we set $X_i = \phi^{-1}(S_i)$ and $Y_i = \psi^{-1}(S_i)$, then the products $X_i \times_S Y_i$ exist, by the above; but then the products $X_i \times_S Y_j$ also exist (3.2.5), and Lemma (3.2.6.4) therefore shows that $X \times_S Y$ exists. $\square$

I | 108

Corollary (3.2.7). — *Let $Z = X \times_S Y$ be the product of two S -preschemes, p and q the projections from Z to X and to Y (respectively), and ϕ (resp. ψ) the structure morphism of X (resp. Y). Let S' be an open subset of S , and U (resp. V) an open subset of X (resp. Y) contained in $\phi^{-1}(S')$ (resp. $\psi^{-1}(S')$). Then the product $U \times_{S'} V$ is canonically identified with the prescheme induced on Z by $p^{-1}(U) \cap q^{-1}(V)$ (considered as an S' -prescheme). In addition, if $f : T \rightarrow X$ and $g : T \rightarrow Y$ are S -morphisms such that $f(T) \subset U$ and $g(T) \subset V$, then the S' -morphism $(f, g)_{S'}$ can be identified with $(f, g)_S$ regarded as a morphism into $p^{-1}(U) \cap q^{-1}(V)$.*

Proof. This follows from Corollary (3.2.5) and Lemma (3.2.6.1). $\square$

(3.2.8). Let (X_α) and (Y_λ) be families of S -preschemes, and X (resp. Y) the sum of the family (X_α) (resp. (Y_λ)) (3.1). Then $X \times_S Y$ can be identified with the sum of the family $(X_\alpha \times_S Y_\lambda)$; this follows immediately from Lemma (3.2.6.3).

(3.2.9).² It follows from (1.8.1) that we can state (3.2.2) in the following manner: $Z = \text{Spec}(B \otimes_A C)$ is not only a product of $X = \text{Spec}(B)$ and $Y = \text{Spec}(C)$ in the category of S -preschemes, but also in the category of locally ringed spaces over S (with a definition of S -morphisms modelled on that of (2.5.2)). The proof of (3.2.6) also proves that, for any two S -preschemes X and Y , the prescheme $X \times_S Y$ is not only the product of X and Y in the category of S -preschemes, but also in the category of locally ringed spaces over the prescheme S .

II | 221

²[Trans.] (3.2.9) is from the errata of EGA II, on page 221, whence the change in page numbering.

3.3. Formal properties of the product; change of the base prescheme

(3.3.1). The reader will notice that all the properties stated in this section, except (3.3.13) and (3.3.15), are true without modification in *any category*, whenever the products involved in the statements *exist* (since it is clear that the notions of an S -object and of an S -morphism can be defined exactly as in (2.5) for any object S of the category).

(3.3.2). First of all, $X \times_S Y$ is a *covariant bifunctor* in X and Y on the category of S -preschemes: it suffices in fact to note that the diagram

$$\begin{array}{ccccc} X \times Y & \xrightarrow{f \times 1} & X' \times Y & \xrightarrow{f' \times 1} & X'' \times Y \\ \downarrow & & \downarrow & & \downarrow \\ X & \xrightarrow{f} & X' & \xrightarrow{f'} & X'' \end{array}$$

is commutative.

Proposition (3.3.3). — *For each S -prescheme X , the first (resp. second) projection from $X \times_S S$ (resp. $S \times_S X$) is a functorial isomorphism from $X \times_S S$ (resp. $S \times_S X$) to X , whose inverse isomorphism is $(1_X, \phi)_S$ (resp. $(\phi, 1_X)_S$), where we denote by ϕ the structure morphism $X \rightarrow S$; we can therefore write, up to a canonical isomorphism,*

$$X \times_S S = S \times_S X = X.$$

Proof. It suffices to prove that the triple $(X, 1_X, \phi)$ is a product of X and S . If T is an S -prescheme, then the only S -morphism from T to S is necessarily the structure morphism $\psi : T \rightarrow S$. If f is an S -morphism from T to X , we necessarily have $\psi = \phi \circ f$, hence our claim. $\square$

Corollary (3.3.4). — *Let X and Y be S -preschemes, with structure morphisms $\phi : X \rightarrow S$ and $\psi : Y \rightarrow S$. If we canonically identify X with $X \times_S S$, and Y with $S \times_S Y$, then the projections $X \times_S Y \rightarrow X$ and $X \times_S Y \rightarrow Y$ are identified with $1_X \times \psi$ and $\phi \times 1_Y$ (respectively).*

The proof is immediate and is left to the reader.

(3.3.5). We can define, in a manner similar to (3.2), the product of a finite number n of S -preschemes, and the existence of these products follows from (3.2.6) by induction on n , and by noting that $(X_1 \times_S X_2 \times_S \cdots \times_S X_{n-1}) \times_S X_n$ satisfies the definition of a product. The uniqueness of the product implies, as in any category, its *commutativity* and *associativity* properties. If, for example, p_1, p_2 , and p_3 denote the projections from $X_1 \times_S X_2 \times_S X_3$, and if we identify this prescheme with $(X_1 \times_S X_2) \times_S X_3$, then the projection to $X_1 \times_S X_2$ is identified with $(p_1, p_2)_S$. I | 109

(3.3.6). Let S and S' be preschemes, and $\phi : S' \rightarrow S$ a morphism, which lets us consider S' as an S -prescheme. For each S -prescheme X , consider the product $X \times_S S'$, and let p and π' be the projections to X and to S' (respectively). Equipped with π' , this product is an S' -prescheme; when we consider it as such, we denote it by $X_{(S')}$ or $X_{(\phi)}$, and we say that this is the prescheme obtained by *base change* (or a *change of base*) from S to S' by means of the morphism ϕ , or the *inverse image* of X by ϕ . We note that, if π is the structure morphism of X , and θ the structure morphism of $X \times_S S'$, considered as an S -prescheme, then the diagram

$$\begin{array}{ccc} X & \xleftarrow{p} & X_{(S')} \\ \pi \downarrow & \theta \swarrow & \downarrow \pi' \\ S & \xleftarrow{\phi} & S' \end{array}$$

is commutative.

(3.3.7). With the notation of (3.3.6), for each S -morphism $f : X \rightarrow Y$, we denote by $f_{(S')}$ the S' -morphism $f \times_S 1 : X_{(S')} \rightarrow Y_{(S')}$, and we say that $f_{(S')}$ is the *base change* (or *inverse image*) of f by ϕ . Therefore $X_{(S')}$ is a *covariant functor* in X , from the category of S -preschemes to that of S' -preschemes.

(3.3.8). The prescheme $X_{(S')}$ can be considered as a solution to a *universal mapping problem*: each S' -prescheme T is also an S -prescheme via ϕ ; each S -morphism $g : T \rightarrow X$ is then uniquely written as $g = p \circ f$, where f is an S' -morphism $T \rightarrow X_{(S')}$, as follows from the definition of the product applied to the S -morphisms g^3 and $\psi : T \rightarrow S'$ (the structure morphism of T).

Proposition (3.3.9). — (“Transitivity of base change”). *Let S'' be a prescheme, and $\phi' : S'' \rightarrow S'$ a morphism. For each S -prescheme X , there exists a canonical functorial isomorphism from the S'' -prescheme $(X_{(\phi)})_{(\phi')}$ to the S'' -prescheme $X_{(\phi \circ \phi')}$.*

Proof. Let T be a S'' -prescheme, ψ its structure morphism, and g an S -morphism from T to X (T being considered as an S -prescheme with structure morphism $\phi \circ \phi' \circ \psi$). Since T is also an S' -prescheme with structure morphism $\phi' \circ \psi$, we can write $g = p \circ g'$, where g' is an S' -morphism $T \rightarrow X_{(\phi)}$, and then $g' = p' \circ g''$, where g'' is an S'' -morphism $T \rightarrow (X_{(\phi)})_{(\phi')}$:

$$\begin{array}{ccccc} X & \xleftarrow{p} & X_{(\phi)} & \xleftarrow{p'} & (X_{(\phi)})_{(\phi')} \\ \pi \downarrow & & \pi' \downarrow & & \downarrow \pi'' \\ S & \xleftarrow{\phi} & S' & \xleftarrow{\phi'} & S'' \end{array}$$

So the result follows by the uniqueness of the solution to a universal mapping problem. $\square$ I | 110

This result can be written as the equality (up to a canonical isomorphism) $(X_{(S')})_{(S'')} = X_{(S'')}$ (if there is no chance of confusion), or also as

$$(3.3.9.1) \quad (X \times_S S') \times_{S'} S'' = X \times_S S'';$$

the functorial nature of the isomorphism defined in (3.3.9) can similarly be expressed by the transitivity formula for base change morphisms

$$(3.3.9.2) \quad (f_{(S')})_{(S'')} = f_{(S'')}$$

for each S -morphism $f : X \rightarrow Y$.

Corollary (3.3.10). — *If X and Y are S -preschemes, then there exists a canonical functorial isomorphism from the S' -prescheme $X_{(S')} \times_{S'} Y_{(S')}$ to the S' -prescheme $(X \times_S Y)_{(S')}$.*

Proof. We have, up to canonical isomorphism,

$$(X \times_S S') \times_{S'} (Y \times_S S') = X \times_S (Y \times_S S') = (X \times_S Y) \times_S S'$$

according to (3.3.9.1) and the associativity of products of S -preschemes. $\square$

The functorial nature of the isomorphism defined in Corollary (3.3.10) can be expressed by the formula

$$(3.3.10.1) \quad (u_{(S')}, v_{(S')})_{S'} = ((u, v)_S)_{(S')}$$

for each pair of S -morphisms $u : T \rightarrow X, v : T \rightarrow Y$.

In other words, the base change functor $X_{(S')}$ commutes with products; it also commutes with sums (3.2.8).

Corollary (3.3.11). — *Let Y be an S -prescheme, and $f : X \rightarrow Y$ a morphism which makes X a Y -prescheme (and, as a result, also an S -prescheme). The prescheme $X_{(S')}$ is then identified with the product $X \times_Y Y_{(S')}$, the projection $X \times_Y Y_{(S')} \rightarrow Y_{(S')}$ being identified with $f_{(S')}$.*

Proof. Let $\psi : Y \rightarrow S$ be the structure morphism of Y ; we have the commutative diagram

$$\begin{array}{ccccc} S' & \xleftarrow{\psi_{(S')}} & Y_{(S')} & \xleftarrow{f_{(S')}} & X_{(S')} \\ \downarrow & & \downarrow & & \downarrow \\ S & \xleftarrow{\psi} & Y & \xleftarrow{f} & X \end{array}$$

³[Trans.] The French text prints f here; the defining pair is (g, ψ) .

We have that $Y_{(S')}$ is identified with $S'_{(\psi)}$, and $X_{(S')}$ with $S'_{(\psi \circ f)}$; taking (3.3.9) and (3.3.4) into account, we thus deduce the corollary. $\square$

(3.3.12). Let $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ be S -morphisms which are *monomorphisms* of preschemes (T, I, 1.1); then $f \times_S g$ is a *monomorphism*. Indeed, if p and q are the projections of $X \times_S Y$, p' and q' the projections of $X' \times_S Y'$, and u and v are S -morphisms $T \rightarrow X \times_S Y$, then the equation $(f \times_S g) \circ u = (f \times_S g) \circ v$ implies that $p' \circ (f \times_S g) \circ u = p' \circ (f \times_S g) \circ v$, or, in other words, that $f \circ p \circ u = f \circ p \circ v$, and since f is a monomorphism, $p \circ u = p \circ v$; using the fact that g is a monomorphism, we similarly obtain $q \circ u = q \circ v$, hence $u = v$.

It follows that, for each base change $S' \rightarrow S$,

$$f_{(S')} : X_{(S')} \longrightarrow X'_{(S')}$$

I | 111

is a monomorphism.⁴

(3.3.13). Let S and S' be affine schemes of rings A and A' (respectively); a morphism $S' \rightarrow S$ then corresponds to a ring homomorphism $A \rightarrow A'$. If X is an S -prescheme, we denote by $X_{(A')}$ or $X \otimes_A A'$ the S' -prescheme $X_{(S')}$; when X is also affine of ring B , $X_{(A')}$ is affine of ring $B_{(A')} = B \otimes_A A'$ obtained by extension of scalars from the A -algebra B to A' .

(3.3.14). With the notation of (3.3.6), for each S -morphism $f : S' \rightarrow X$, we have that $f' = (f, 1_{S'})_S$ is an S' -morphism $S' \rightarrow X' = X_{(S')}$ such that $p \circ f' = f$, $\pi' \circ f' = 1_{S'}$, or, in other words, an S' -section of X' ; conversely, if f' is such an S' -section, then $f = p \circ f'$ is an S -morphism $S' \rightarrow X$. We thus define a canonical *bijective correspondence*

$$\mathrm{Hom}_S(S', X) \simeq \mathrm{Hom}_{S'}(S', X').$$

We say that f' is the *graph morphism* of f , and we denote it by Γ_f .

(3.3.15). Given a prescheme X , which we can always consider as a $\mathbf{Z}$ -prescheme, it follows, in particular, from (3.3.14) that the X -sections of $X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ (where T is an indeterminate) correspond bijectively with *morphisms* $X \rightarrow \mathbf{Z}[T]$ ⁵. We will show that these X -sections also correspond bijectively with *sections of the structure sheaf* $\mathcal{O}_X$ over X . Indeed, let (U_α) be a cover of X by affine open subsets; let $u : X \rightarrow X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ be an X -morphism, and let u_α be its restriction to U_α ; if A_α is the ring of the affine scheme U_α , then $U_\alpha \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ is an affine scheme of ring $A_\alpha[T]$ (3.2.2), and u_α canonically corresponds to an A_α -homomorphism $A_\alpha[T] \rightarrow A_\alpha$ (1.7.3). Now, since such a homomorphism is completely determined by the data of the image of T in A_α , let $s_\alpha \in A_\alpha = \Gamma(U_\alpha, \mathcal{O}_X)$, and if we suppose that the restrictions of u_α and u_β to an affine open subset $V \subset U_\alpha \cap U_\beta$ coincide, then we see immediately that s_α and s_β coincide on V ; thus the family (s_α) consists of the restrictions to U_α of a section s of $\mathcal{O}_X$ over X ; conversely, it is clear that such a section defines a family (u_α) of morphisms which are the restrictions to U_α of an X -morphism $X \rightarrow X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$. This result is generalized in (II, 1.7.12).

⁴[Trans.] The French formula prints $Y_{(S')}$; since 3.3.12 assumes $f : X \rightarrow X'$, the base-changed codomain is $X'_{(S')}$.

⁵[Trans.] The French text prints $\mathbf{Z}[T] \rightarrow X$; Proposition (3.3.14) and the ensuing local ring homomorphisms $A_\alpha[T] \rightarrow A_\alpha$ require the scheme morphism $X \rightarrow \mathbf{Z}[T]$.

3.4. Points of a prescheme with values in a prescheme; geometric points

(3.4.1). Let X be a prescheme; for each prescheme T , we then denote by $X(T)$ the set $\text{Hom}(T, X)$ of morphisms $T \rightarrow X$, and the elements of this set are called *the points of X with values in T* . If we associate to each morphism $f : T \rightarrow T'$ the map $u' \mapsto u' \circ f$ from $X(T')$ to $X(T)$, we see, for fixed X , that $X(T)$ is a *contravariant functor in T* , from the category of preschemes to that of sets. In addition, each morphism of preschemes $g : X \rightarrow Y$ defines a functorial homomorphism $X(T) \rightarrow Y(T)$, which sends $v \in X(T)$ to $g \circ v$.

(3.4.2). Given sets P, Q , and R , and maps $\phi : P \rightarrow R$ and $\psi : Q \rightarrow R$, we define the *fibre product of P and Q over R* (with respect to ϕ and ψ) as the subset of the product set $P \times Q$ consisting of the pairs (p, q) such that $\phi(p) = \psi(q)$; we denote it by $P \times_R Q$. Definition (3.2.1) of the product of S -preschemes can be interpreted, with the notation of (3.4.1), via the formula

I | 112

$$(3.4.2.1) \quad (X \times_S Y)(T) = X(T) \times_{S(T)} Y(T),$$

where the maps $X(T) \rightarrow S(T)$ and $Y(T) \rightarrow S(T)$ correspond to the structure morphisms $X \rightarrow S$ and $Y \rightarrow S$.

(3.4.3). If we are given a prescheme S and we consider only the S -preschemes and S -morphisms, then we will denote by $X(T)_S$ the set $\text{Hom}_S(T, X)$ of S -morphisms $T \rightarrow X$, and suppress the subscript S when there is no chance of confusion; we say that the elements of $X(T)_S$ are the *points* (or *S -points*, when there is a possibility of confusion) of the S -prescheme X with values in the S -prescheme T . In particular, an S -section of X is none other than a *point of X with values in S* . The formula (3.4.2.1) can then be written as

$$(3.4.3.1) \quad (X \times_S Y)(T)_S = X(T)_S \times Y(T)_S;$$

more generally, if Z is an S -prescheme, and X, Y , and T are Z -preschemes (thus *ipso facto* S -preschemes), then we have

$$(3.4.3.2) \quad (X \times_Z Y)(T)_S = X(T)_S \times_{Z(T)_S} Y(T)_S.$$

We note that, to show that a triple (W, r, s) consisting of an S -prescheme W and S -morphisms $r : W \rightarrow X$ and $s : W \rightarrow Y$ is a product of X and Y (over Z), it suffices, by definition, to check that, for each S -prescheme T , the diagram

$$\begin{array}{ccc} W(T)_S & \xrightarrow{r'} & X(T)_S \\ s' \downarrow & & \downarrow \phi' \\ Y(T)_S & \xrightarrow{\psi'} & Z(T)_S \end{array}$$

makes $W(T)_S$ the fibre product of $X(T)_S$ and $Y(T)_S$ over $Z(T)_S$, where r' and s' correspond to r and s , and ϕ' and ψ' to the structure morphisms $\phi : X \rightarrow Z$ and $\psi : Y \rightarrow Z$.

(3.4.4). When T (resp. S) in the above is an affine scheme of ring B (resp. A), we replace T (resp. S) by B (resp. A) in the above notation, and we then call the elements of $X(B)$ the *points of X with values in the ring B* , and the elements of $X(B)_A$ the *points of the A -prescheme X with values in the A -algebra B* . We note that $X(B)$ and $X(B)_A$ are *covariant* functors in B . We similarly write $X(T)_A$ for the set of points of the A -prescheme X with values in the A -prescheme T .

(3.4.5). Consider, in particular, the case where T is of the form $\text{Spec}(A)$, where A is a *local* ring; the elements of $X(A)$ then correspond bijectively to *local* homomorphisms $\mathcal{O}_x \rightarrow A$ for $x \in X$ (2.4.4); we say that the point x of the underlying space of X is the *location*⁶ of the point of X with values in A to which it corresponds.

More specifically, we define the *geometric points* of a prescheme X to be the *points of X with values in a field K* : the data of such a point is equivalent to the data of its location x in the underlying space of X , and of an *extension* K of $k(x)$; K will be called the *field of values* of the corresponding geometric

I | 113

⁶[Trans.] We also say that the geometric point lies over this x .

point, and we say that this geometric point is *located at* x . We thus define a map $X(K) \rightarrow X$, sending a geometric point with values in K to its location.

If $S' = \text{Spec}(K)$ is an S -prescheme (in other words, if K is considered as an extension of the residue field $k(s)$, where $s \in S$), and if X is an S -prescheme, then an element of $X(K)_S$, or, as we say, a *geometric point of X lying over s with values in K* , consists of the data of a $k(s)$ -monomorphism from the residue field $k(x)$ to K , where x is a point of X lying over s (therefore $k(x)$ is an extension of $k(s)$).

In particular, if $S = \text{Spec}(K) = \{\xi\}$, then the geometric points of X with values in K can be identified with the points $x \in X$ such that $k(x) = K$; we say that these latter points are the K -rational points of the K -prescheme X ; if K' is an extension of K , then the geometric points of X with values in K' bijectively correspond to the K' -rational points of $X' = X_{(K')}$ (3.3.14).

Lemma (3.4.6). — Let X_i ($1 \leq i \leq n$) be S -preschemes, s a point of S , and x_i ($1 \leq i \leq n$) points of X_i lying over s . Then there exists an extension K of $k(s)$ and a geometric point of the product $Y = X_1 \times_S X_2 \times_S \cdots \times_S X_n$, with values in K , whose projections to the X_i are localized at the x_i .

Proof. There exist $k(s)$ -monomorphisms $k(x_i) \rightarrow K$, all in the same extension K of $k(s)$ (Bourbaki, Alg., chap. V, §4, prop. 2). The compositions $k(s) \rightarrow k(x_i) \rightarrow K$ are all identical, and so the morphisms $\text{Spec}(K) \rightarrow X_i$ corresponding to the $k(x_i) \rightarrow K$ are all S -morphisms, and we thus conclude that they define a unique morphism $\text{Spec}(K) \rightarrow Y$. If y is the corresponding point of Y , it is clear that its projection to each of the X_i is x_i . $\square$

Proposition (3.4.7). — Let X_i ($1 \leq i \leq n$) be S -preschemes, and, for each index i , let x_i be a point of X_i . For there to exist a point y of $Y = X_1 \times_S X_2 \times \cdots \times_S X_n$ whose image is x_i under the i th projection for each $1 \leq i \leq n$, it is necessary and sufficient that the x_i all lie above the same point s of S .

Proof. The condition is evidently necessary; Lemma (3.4.6) proves that it is sufficient. $\square$

In other words, if we denote by (X) the underlying set of X , we see that we have a canonical surjective function $(X \times_S Y) \rightarrow (X) \times_{(S)} (Y)$; we must point out that this function is *not injective* in general; in other words, there can exist multiple distinct points z in $X \times_S Y$ that have the same projections $x \in X$ and $y \in Y$; we have already seen this when S , X , and Y are prime spectra of fields k , K , and K' (respectively), since the tensor product $K \otimes_k K'$ has, in general, multiple distinct prime ideals (cf. (3.4.9)).

Corollary (3.4.8). — Let $f : X \rightarrow Y$ be an S -morphism, and $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ the S' -morphism obtained from f by the base extension $S' \rightarrow S$. Let p (resp. q) be the projection $X_{(S')} \rightarrow X$ (resp. $Y_{(S')} \rightarrow Y$); for every subset M of X , we have

$$q^{-1}(f(M)) = f_{(S')}(p^{-1}(M)).$$

I | 114

Proof. Indeed, by (3.3.11), $X_{(S')}$ can be identified with the product $X \times_Y Y_{(S')}$ thanks to the commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{p} & X_{(S')} \\ f \downarrow & & \downarrow f_{(S')} \\ Y & \xleftarrow{q} & Y_{(S')} \end{array}$$

By (3.4.7), the equation $q(y') = f(x)$ for $x \in M$ and $y' \in Y_{(S')}$ is equivalent to the existence of some $x' \in X_{(S')}$ such that $p(x') = x$ and $f_{(S')}(x') = y'$, whence the corollary. $\square$

Lemma (3.4.6) can be made clearer in the following manner:

Proposition (3.4.9). — Let X and Y be S -preschemes, x a point of X , and y a point of Y , with both x and y lying above the same point $s \in S$. The set of points of $X \times_S Y$ with projections x and y is in canonical bijective correspondence with the set of types of composite extensions of $k(x)$ and $k(y)$ considered as extensions of $k(s)$ (Bourbaki, Alg., chap. VIII, §8, prop. 2).

Proof. Let p (resp. q) be the projection from $X \times_S Y$ to X (resp. Y), and E the subspace $p^{-1}(x) \cap q^{-1}(y)$ of the underlying space of $X \times_S Y$. First, note that the morphisms $\text{Spec}(k(x)) \rightarrow S$ and $\text{Spec}(k(y)) \rightarrow S$ factor as $\text{Spec}(k(x)) \rightarrow \text{Spec}(k(s)) \rightarrow S$ and $\text{Spec}(k(y)) \rightarrow \text{Spec}(k(s)) \rightarrow S$; since $\text{Spec}(k(s)) \rightarrow S$ is a monomorphism (2.4.7), it follows from (3.2.4) that we have

$$P = \text{Spec}(k(x)) \times_S \text{Spec}(k(y)) = \text{Spec}(k(x)) \times_{\text{Spec}(k(s))} \text{Spec}(k(y)) = \text{Spec}(k(x) \otimes_{k(s)} k(y)).$$

We will define two maps, $\alpha : P_0 \rightarrow E$ and $\beta : E \rightarrow P_0$, inverse to one another (where P_0 denotes the underlying set of the prescheme P). If $i : \text{Spec}(k(x)) \rightarrow X$ and $j : \text{Spec}(k(y)) \rightarrow Y$ are the canonical morphisms (2.4.5), we take α to be the map of underlying spaces corresponding to the morphism $i \times_S j$. On the other hand, every $z \in E$ defines, by hypothesis, two $k(s)$ -monomorphisms, $k(x) \rightarrow k(z)$ and $k(y) \rightarrow k(z)$, and thus a $k(s)$ -homomorphism⁷ $k(x) \otimes_{k(s)} k(y) \rightarrow k(z)$, and thus a morphism $\text{Spec}(k(z)) \rightarrow P$; $\beta(z)$ will be the image of z in P_0 under this morphism. The verification of the fact that $\alpha \circ \beta$ and $\beta \circ \alpha$ are the identity maps follows from (2.4.5) and the definition of the product (3.2.1). Finally, we know that P_0 is in bijective correspondence with the set of types of composite extensions of $k(x)$ and $k(y)$ (Bourbaki, *Alg.*, chap. VIII, §8, prop. 1). $\square$

3.5. Surjections and injections

(3.5.1). In a general sense, consider a property **P** of morphisms of preschemes, and the following two propositions:

- (i) If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are S -morphisms that have property **P**, then $f \times_S g$ also has property **P**.
- (ii) If $f : X \rightarrow Y$ is an S -morphism that has property **P**, then every S' -morphism $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ obtained from f by a base extension $S' \rightarrow S$ also has property **P**.

Since $f_{(S')} = f \times_S 1_{S'}$, we see that, if, for every prescheme X , the identity 1_X has property **P**, then (i) implies (ii); on the other hand, since $f \times_S g$ is the composite morphism

$$X \times_S Y \xrightarrow{f \times 1_Y} X' \times_S Y \xrightarrow{1_{X'} \times g} X' \times_S Y',$$

we see that, if the *composite* of two morphisms having property **P** also has property **P**, then so does the product $f \times_S g$, and so (ii) implies (i).

A first application of this remark is

Proposition (3.5.2). —

- (i) If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are surjective S -morphisms, then $f \times_S g$ is surjective.
- (ii) If $f : X \rightarrow Y$ is a surjective S -morphism, then $f_{(S')}$ is surjective for every base extension $S' \rightarrow S$.

Proof. The composition of any two surjections being a surjection, it suffices to prove (ii); but this proposition follows immediately from (3.4.8) applied to $M = X$. $\square$

Proposition (3.5.3). — For a morphism $f : X \rightarrow Y$ to be surjective, it is necessary and sufficient that, for every field K and every morphism $\text{Spec}(K) \rightarrow Y$, there exist an extension K' of K and a morphism $\text{Spec}(K') \rightarrow X$ that make the following diagram commute:

$$\begin{array}{ccc} X & \longleftarrow & \text{Spec}(K') \\ f \downarrow & & \downarrow \\ Y & \longleftarrow & \text{Spec}(K). \end{array}$$

Proof. The condition is sufficient because, for all $y \in Y$, it suffices to apply it to a morphism $\text{Spec}(K) \rightarrow Y$ corresponding to a monomorphism $k(y) \rightarrow K$, with K being an extension of $k(y)$ (2.4.6). Conversely, suppose that f is surjective, and let $y \in Y$ be the image of the unique point of $\text{Spec}(K)$; there exists some $x \in X$ such that $f(x) = y$; we will consider the corresponding monomorphism $k(y) \rightarrow k(x)$ (2.2.1); it then suffices to take K' to be an extension of $k(y)$ such that there exist

⁷[Trans.] The French text prints “monomorphism” here. The induced map from the tensor product need not be injective; only a homomorphism is needed to define the ensuing morphism of spectra.

$k(y)$ -monomorphisms from $k(x)$ and K to K' (Bourbaki, *Alg.*, chap. V, §4, prop. 2); the morphism $\text{Spec}(K') \rightarrow X$ corresponding to $k(x) \rightarrow K'$ is exactly that for which we are searching. $\square$

With the language introduced in (3.4.5), we can further say that *every geometric point of Y with values in K comes from a geometric point of X with values in an extension of K .*

Definition (3.5.4). — We say that a morphism $f : X \rightarrow Y$ of preschemes is *universally injective*, or a *radicial morphism*, if, for every field K , the corresponding map $X(K) \rightarrow Y(K)$ is injective.

It follows immediately from the definitions that every *monomorphism of preschemes* (T, 1.1) is radicial.

(3.5.5). For a morphism $f : X \rightarrow Y$ to be radicial, it suffices that the condition of Definition (3.5.4) hold for every *algebraically closed* field. In fact, if K is an arbitrary field, and K' an algebraically closed extension of K , then the diagram

$$\begin{array}{ccc} X(K) & \xrightarrow{\alpha} & Y(K) \\ \phi \downarrow & & \downarrow \phi' \\ X(K') & \xrightarrow{\alpha'} & Y(K') \end{array}$$

commutes, where ϕ and ϕ' come from the morphism $\text{Spec}(K') \rightarrow \text{Spec}(K)$, and α and α' correspond to f . However, ϕ is injective, and so too is α' , by hypothesis; hence α is necessarily injective.

Proposition (3.5.6). — *Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms of preschemes.*

- (i) *If f and g are radicial, then so is $g \circ f$.*
- (ii) *Conversely, if $g \circ f$ is radicial, then so is f .*

Proof. Taking into account Definition (3.5.4), the proposition reduces to the corresponding claims for the maps $X(K) \rightarrow Y(K) \rightarrow Z(K)$, and these claims are evident. $\square$

Proposition (3.5.7). —

- (i) *If the S -morphisms $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are radicial, then so is $f \times_S g$.*
- (ii) *If the S -morphism $f : X \rightarrow Y$ is radicial, then so is $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ for every extension $S' \rightarrow S$ of the base prescheme.*

Proof. Given (3.5.1), it suffices to prove (i). We have seen in (3.4.2.1) that

$$\begin{aligned} (X \times_S Y)(K) &= X(K) \times_{S(K)} Y(K), \\ (X' \times_S Y')(K) &= X'(K) \times_{S(K)} Y'(K), \end{aligned}$$

with the map $(X \times_S Y)(K) \rightarrow (X' \times_S Y')(K)$ corresponding to $f \times_S g$ thus being identified with $(u, v) \rightarrow (f \circ u, g \circ v)$, and the proposition follows immediately. $\square$

Proposition (3.5.8). — *For a morphism $f = (\psi, \theta) : X \rightarrow Y$ to be radicial, it is necessary and sufficient for ψ to be injective and for the monomorphism $\theta^x : k(\psi(x)) \rightarrow k(x)$ to make $k(x)$ a radicial extension of $k(\psi(x))$ for every $x \in X$.*

Proof. We suppose that f is radicial and first show that the equation $\psi(x_1) = \psi(x_2) = y$ necessarily implies that $x_1 = x_2$. Indeed, there exists a field K extending $k(y)$, together with $k(y)$ -monomorphisms $k(x_1) \rightarrow K$ and $k(x_2) \rightarrow K$ (Bourbaki, *Alg.*, chap. V, §4, prop. 2); the corresponding morphisms $u_1 : \text{Spec}(K) \rightarrow X$ and $u_2 : \text{Spec}(K) \rightarrow X$ are then such that $f \circ u_1 = f \circ u_2$, and so $u_1 = u_2$ by hypothesis, and this implies, in particular, that $x_1 = x_2$. We now consider $k(x)$ as an extension of $k(\psi(x))$ by means of θ^x : if $k(x)$ is not a radicial extension of $k(\psi(x))$, then there exist two distinct $k(\psi(x))$ -monomorphisms from $k(x)$ to an algebraically closed extension K of $k(\psi(x))$, and the two corresponding morphisms $\text{Spec}(K) \rightarrow X$ would contradict the hypothesis. Conversely, taking (2.4.6) into account, it is immediate that the conditions stated are sufficient for f to be radicial. $\square$

Corollary (3.5.9). — *If A is a ring, and S is a multiplicative set of A , then the canonical morphism $\text{Spec}(S^{-1}A) \rightarrow \text{Spec}(A)$ is radicial.*

Proof. Indeed, this morphism is a monomorphism (1.6.2). $\square$

Corollary (3.5.10). — Let $f : X \rightarrow Y$ be a radicial morphism, $g : Y' \rightarrow Y$ a morphism, and $X' = X_{(Y')} = X \times_Y Y'$. Then the radicial morphism $f_{(Y')}$ (3.5.7, ii) is a bijection from the underlying space of X' to $g^{-1}(f(X))$; further, for every field K , the set $X'(K)$ can be identified with the inverse image of the subset $X(K) \subseteq Y(K)$ under the map $Y'(K) \rightarrow Y(K)$ corresponding to g .

Proof. The first claim follows immediately from (3.5.8) and (3.4.8); the second, from the commutativity of the following diagram:

$$\begin{array}{ccc} X'(K) & \longrightarrow & Y'(K) \\ \downarrow & & \downarrow \\ X(K) & \longrightarrow & Y(K) \end{array}$$

$\square$

Remark (3.5.11). — We say that a morphism $f = (\psi, \theta)$ of preschemes is *injective* if the map ψ is injective. For a morphism $f = (\psi, \theta) : X \rightarrow Y$ to be radicial, it is necessary and sufficient that, for every morphism $Y' \rightarrow Y$, the morphism $f_{(Y')} : X_{(Y')} \rightarrow Y'$ be injective (which justifies the terminology of a *universally injective* morphism). In fact, the condition is necessary by (3.5.7, ii) and (3.5.8). Conversely, the condition first implies that ψ is injective; if, for some $x \in X$, the monomorphism $\theta^x : k(\psi(x)) \rightarrow k(x)$ were not radicial, then there would be an extension K of $k(\psi(x))$, and two distinct morphisms $\text{Spec}(K) \rightarrow X$ corresponding to the same morphism $\text{Spec}(K) \rightarrow Y$ (3.5.8). But then, setting $Y' = \text{Spec}(K)$, there would be two distinct Y' -sections of $X_{(Y')}$ (3.3.14), which contradicts the hypothesis that $f_{(Y')}$ is injective.

3.6. Fibres

Proposition (3.6.1). — Let $f : X \rightarrow Y$ be a morphism, y a point of Y , and $\mathfrak{a}_y$ an ideal of definition of $\mathcal{O}_y$ for the $\mathfrak{m}_y$ -preadic topology. Then the projection $p : X \times_Y \text{Spec}(\mathcal{O}_y/\mathfrak{a}_y) \rightarrow X$ is a homeomorphism from the underlying space of the prescheme $X \times_Y \text{Spec}(\mathcal{O}_y/\mathfrak{a}_y)$ to the fibre $f^{-1}(y)$ equipped with the topology induced from that of the underlying space of X .

Proof. Since $\text{Spec}(\mathcal{O}_y/\mathfrak{a}_y) \rightarrow Y$ is radicial ((3.5.4) and (2.4.7)), since $\text{Spec}(\mathcal{O}_y/\mathfrak{a}_y)$ is a single point, and since the ideal $\mathfrak{m}_y/\mathfrak{a}_y$ is nilpotent by hypothesis (1.1.12), we already know ((3.5.10) and (3.3.4)) that p identifies, as sets, the underlying space of $X \times_Y \text{Spec}(\mathcal{O}_y/\mathfrak{a}_y)$ with $f^{-1}(y)$; everything reduces to proving that p is a homeomorphism. By (3.2.7), the question is local on X and Y , and so we can suppose that $X = \text{Spec}(B)$ and $Y = \text{Spec}(A)$, with B being an A -algebra. The morphism p then corresponds to the homomorphism $1 \otimes \phi : B \rightarrow B \otimes_A A'$, where $A' = A_y/\mathfrak{a}_y$ and ϕ is the canonical map from A to A' . Then every element of $B \otimes_A A'$ can be written as

$$\sum_i b_i \otimes \phi(a_i) / \phi(s) = \left(\sum_i (a_i b_i \otimes 1) \right) (1 \otimes \phi(s))^{-1},$$

where $s \notin \mathfrak{j}_y$, and Proposition (1.2.4) applies. $\square$

(3.6.2). Throughout the rest of this treatise, whenever we consider a fibre $f^{-1}(y)$ of a morphism as having the structure of a $k(y)$ -prescheme, it will always be the prescheme obtained by transporting the structure of $X \times_Y \text{Spec}(k(y))$ by the projection to X . We will also write this latter product as $X \times_Y k(y)$, or $X \otimes_{\mathcal{O}_Y} k(y)$; more generally, if B is an $\mathcal{O}_Y$ -algebra, we will denote by $X \times_Y B$ or $X \otimes_{\mathcal{O}_Y} B$ the product $X \times_Y \text{Spec}(B)$.

With the preceding convention, it follows from (3.5.10) that the points of X with values in an extension K of $k(y)$ are identified with the points of $f^{-1}(y)$ with values in K .

(3.6.3). Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms, and $h = g \circ f$ their composition; for all $z \in Z$, the fibre $h^{-1}(z)$ is a prescheme isomorphic to

$$X \times_Z \text{Spec}(k(z)) = (X \times_Y Y) \times_Z \text{Spec}(k(z)) = X \times_Y g^{-1}(z).$$

In particular, if U is an open subset of X , then the prescheme induced on $U \cap f^{-1}(y)$ by the prescheme $f^{-1}(y)$ is isomorphic to $f_U^{-1}(y)$ (f_U being the restriction of f to U). I | 118

Proposition (3.6.4). — (Transitivity of fibres) Let $f : X \rightarrow Y$ and $g : Y' \rightarrow Y$ be morphisms; let $X' = X \times_Y Y' = X_{(Y')}$ and $f' = f_{(Y')} : X' \rightarrow Y'$. For every $y' \in Y'$, if we let $y = g(y')$, then the prescheme $f'^{-1}(y')$ is isomorphic to $f^{-1}(y) \otimes_{k(y)} k(y')$.

Proof. Indeed, it suffices to remark that the two preschemes $(X \otimes_Y k(y)) \otimes_{k(y)} k(y')$ and $(X \times_Y Y') \otimes_{Y'} k(y')$ are both canonically isomorphic to $X \times_Y \text{Spec}(k(y'))$ by (3.3.9.1). $\square$

In particular, if V is an open neighbourhood of y in Y , and we denote by f_V the restriction of f to the induced prescheme on $f^{-1}(V)$, then the preschemes $f^{-1}(y)$ and $f_V^{-1}(y)$ are canonically identified.

Proposition (3.6.5). — Let $f : X \rightarrow Y$ be a morphism, y a point of Y , Z the local prescheme $\text{Spec}(\mathcal{O}_y)$, and $p = (\psi, \theta)$ the projection $X \times_Y Z \rightarrow X$; then p is a homeomorphism from the underlying space of $X \times_Y Z$ to the subspace $f^{-1}(Z)$ of X (when the underlying space of Z is identified with a subspace of Y , cf. (2.4.2)), and, for all $t \in X \times_Y Z$, letting $z = \psi(t)$, we have that $\theta_t^\#$ is an isomorphism from $\mathcal{O}_z$ to $\mathcal{O}_t$.

Proof. Since Z (identified as a subspace of Y) is contained in every affine open containing y (2.4.2), we can, as in (3.6.1), reduce to the case where $X = \text{Spec}(A)$ and $Y = \text{Spec}(B)$ are affine schemes, with A being a B -algebra. Then $X \times_Y Z$ is the prime spectrum of $A \otimes_B B_y$, and this ring is canonically identified with $S^{-1}A$, where S is the image of $B - j_y$ in A (0, 1.5.2); since p then corresponds to the canonical homomorphism $A \rightarrow S^{-1}A$, the proposition follows from (1.6.2). $\square$

3.7. Application: reduction of a prescheme mod. $\mathfrak{J}$ This section, which makes use of notions and results from later in Chapter I and from Chapter II, will not be used in what follows in this treatise, and is only intended for readers familiar with classical algebraic geometry.

(3.7.1). Let A be a ring, X an A -prescheme, and $\mathfrak{J}$ an ideal of A ; then $X_0 = X \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$ -prescheme, which we sometimes say is induced from X by reduction mod. $\mathfrak{J}$.

(3.7.2). This terminology is used especially when A is a local ring and $\mathfrak{J}$ its maximal ideal, so that X_0 is a prescheme over the residue field $k = A/\mathfrak{J}$ of A .

When A is also integral, with field of fractions K , we can consider the K -prescheme $X' = X \otimes_A K$. By an abuse of language which we will not use, it has been customary until now to say that X_0 is induced from X' by reduction mod. $\mathfrak{J}$. In the cases where this language was used, A was a local ring of dimension 1 (most often a discrete valuation ring), and it was understood (more or less explicitly) that the given K -prescheme X' was a closed subprescheme of a K -prescheme P' (in fact, a projective space of type $\mathbf{P}_K^r$, cf. (II, 4.1.1)), itself of the form $P' = P \otimes_A K$, where P is a given A -prescheme (in fact, the A -scheme $\mathbf{P}_A^r$, with the notation of (II, 4.1.1)). In our language, the definition of X_0 in terms of X' is formulated as follows:

We consider the affine scheme $Y = \text{Spec}(A)$, formed of two points, the unique closed point $y = \mathfrak{J}$ and the generic point (0) , the singleton set U of the generic point being thus an open $U = \text{Spec}(K)$ in Y . If X is an A -prescheme (or, in other words, a Y -prescheme), then $X \otimes_A K = X'$ is exactly the prescheme induced by X on $\psi^{-1}(U)$, denoting by ψ the structure morphism $X \rightarrow Y$. In particular, if ϕ is the structure morphism $P \rightarrow Y$, then a closed subprescheme X' of $P' = \phi^{-1}(U)$ is a (locally closed) subprescheme of P . If P is Noetherian (for example, if A is Noetherian and P is of finite type over A), then there exists a smallest closed subprescheme $X = \bar{X}'$ of P containing X' (9.5.10), and X' is the prescheme induced by X on the open $\phi^{-1}(U) \cap X$, and so is isomorphic to $X \otimes_A K$ (9.5.10). The immersion of X' into $P' = P \otimes_A K$ thus lets us canonically consider X' as being of the form $X' = X \otimes_A K$, where X is an A -prescheme. We can then consider the reduced mod. $\mathfrak{J}$ prescheme $X_0 = X \otimes_A k$, which is exactly the fibre $\psi^{-1}(y)$ of the closed point y . Up until now, lacking the adequate terminology, we had avoided explicitly introducing the A -prescheme X . One should note, however, that all the claims normally made about the “reduced mod. $\mathfrak{J}$ ” prescheme X_0 should be seen as consequences of more complete claims about X itself, and cannot be satisfactorily formulated or understood except by interpreting them as such. It seems also that any hypotheses made on X_0 always reduce to hypotheses on X itself (independent of the prior data of an immersion of X' in $\mathbf{P}_K^r$), which lets us give more intrinsic statements.

(3.7.3). Lastly, we draw attention to a very particular fact, which has undoubtedly contributed to slowing the conceptual clarification of the situation envisaged here: if A is a discrete valuation ring, and if X is *proper* over A (which is indeed the case if X is a closed subscheme of some $\mathbf{P}_A^r$, cf. (II, 5.5.4)), then the points of X with values in A and the points of X' with values in K are in bijective correspondence (II, 7.3.8). This is why we have often thought we were proving facts about X' , when in reality we were proving facts about X ; those facts remain valid (in this form) when the base local ring is no longer assumed to have dimension 1.

§4. SUBPRESCHEMES AND IMMERSION MORPHISMS

4.1. Subpreschemes

(4.1.1). As the notion of a quasi-coherent sheaf (0, 5.1.3) is local, a quasi-coherent $\mathcal{O}_X$ -module $\mathcal{F}$ on a prescheme X can be defined by the following condition: for each affine open V of X , $\mathcal{F}|_V$ is isomorphic to the sheaf associated to a $\Gamma(V, \mathcal{O}_X)$ -module (1.4.1). It is clear that, on a prescheme X , the structure sheaf $\mathcal{O}_X$ is quasi-coherent, and that the kernels, cokernels, and images of homomorphisms of quasi-coherent $\mathcal{O}_X$ -modules, as well as inductive limits and direct sums of quasi-coherent $\mathcal{O}_X$ -modules, are also quasi-coherent (Theorem (1.3.7) and Corollary (1.3.9)).

Proposition (4.1.2). — *Let X be a prescheme, and $\mathcal{I}$ a quasi-coherent sheaf of ideals of $\mathcal{O}_X$. Then the support Y of the sheaf $\mathcal{O}_X/\mathcal{I}$ is closed, and if we denote by $\mathcal{O}_Y$ the restriction of $\mathcal{O}_X/\mathcal{I}$ to Y , then $(Y, \mathcal{O}_Y)$ is a prescheme.*

Proof. It evidently suffices (2.1.3) to consider the case where X is an affine scheme, and to show that, in this case, Y is closed in X , and is an *affine scheme*. Indeed, if $X = \text{Spec}(A)$, then we have $\mathcal{O}_X = \tilde{A}$ and $\mathcal{I} = \tilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A (1.4.1); Y is then equal to the closed subset $V(\mathfrak{J})$ of X and can be identified with the prime spectrum of the ring $B = A/\mathfrak{J}$ (1.1.11); in addition, if ϕ is the canonical homomorphism $A \rightarrow B = A/\mathfrak{J}$, then the direct image $\phi_*(\tilde{B})$ is canonically identified with the sheaf $\tilde{A}/\tilde{\mathfrak{J}} = \mathcal{O}_X/\mathcal{I}$ (Proposition (1.6.3) and Corollary (1.3.9)), which finishes the proof. $\square$

We say that $(Y, \mathcal{O}_Y)$ is the *subprescheme* of $(X, \mathcal{O}_X)$ defined by the sheaf of ideals $\mathcal{I}$; this is a particular case of the more general notion of a *subprescheme*:

Definition (4.1.3). — We say that a ringed space $(Y, \mathcal{O}_Y)$ is a subprescheme of a prescheme $(X, \mathcal{O}_X)$ if:

- 1st. Y is a locally closed subspace of X ;
- 2nd. if U denotes the largest open subset of X containing Y such that Y is closed in U (equivalently, the complement in X of the boundary of Y with respect to $\bar{Y}$), then $(Y, \mathcal{O}_Y)$ is a subprescheme of $(U, \mathcal{O}_X|_U)$ defined by a quasi-coherent sheaf of ideals of $\mathcal{O}_X|_U$.

We say that the subprescheme $(Y, \mathcal{O}_Y)$ of $(X, \mathcal{O}_X)$ is closed if Y is closed in X (in which case $U = X$).

It follows immediately from this definition and Proposition (4.1.2) that the closed subpreschemes of X are in canonical *bijective correspondence* with the quasi-coherent sheaves of ideals $\mathcal{I}$ of $\mathcal{O}_X$, since if two such sheaves $\mathcal{I}$ and $\mathcal{I}'$ have the same (closed) support Y , and if the restrictions of $\mathcal{O}_X/\mathcal{I}$ and $\mathcal{O}_X/\mathcal{I}'$ to Y are identical, then we have $\mathcal{I}' = \mathcal{I}$.

(4.1.4). Let $(Y, \mathcal{O}_Y)$ be a subprescheme of X , U the largest open subset of X such that Y is closed (and thus contained) in U , and V an open subset of X contained in U ; then $V \cap Y$ is closed in V . In addition, if Y is defined by the quasi-coherent sheaf of ideals $\mathcal{I}$ of $\mathcal{O}_X|_U$, then $\mathcal{I}|_V$ is a quasi-coherent sheaf of ideals of $\mathcal{O}_X|_V$, and it is immediate that the prescheme induced by Y on $Y \cap V$ is the closed subprescheme of V defined by the sheaf of ideals $\mathcal{I}|_V$. Conversely:

Proposition (4.1.5). — *Let $(Y, \mathcal{O}_Y)$ be a ringed space such that Y is a subspace of X , and suppose there is a cover (V_α) of Y by open subsets of X such that, for each α , $Y \cap V_\alpha$ is closed in V_α , and the ringed space $(Y \cap V_\alpha, \mathcal{O}_Y|_{(Y \cap V_\alpha)})$ is a closed subprescheme of the prescheme induced on V_α by X . Then $(Y, \mathcal{O}_Y)$ is a subprescheme of X .*

Proof. The hypotheses imply that Y is locally closed in X and that the largest open U in which Y is closed (and thus contained) contains all the V_α ; we can thus reduce to the case where $U = X$ and Y

is closed in X . We then define a quasi-coherent sheaf of ideals $\mathcal{J}$ of $\mathcal{O}_X$ by taking $\mathcal{J}|_{V_\alpha}$ to be the sheaf of ideals of $\mathcal{O}_X|_{V_\alpha}$ which defines the closed subprescheme $(Y \cap V_\alpha, \mathcal{O}_Y|_{(Y \cap V_\alpha)})$, and, for each open subset W of X not intersecting Y , $\mathcal{J}|_W = \mathcal{O}_X|_W$. We see immediately, by Definition (4.1.3) and (4.1.4), that there exists a unique sheaf of ideals $\mathcal{J}$ satisfying these conditions, and that it defines the closed subprescheme $(Y, \mathcal{O}_Y)$. $\square$

In particular, the *induced* (by X) prescheme on an open subset of X is a subprescheme of X .

Proposition (4.1.6). — *A subprescheme (resp. a closed subprescheme) of a subprescheme (resp. closed subprescheme) of X is canonically identified with a subprescheme (resp. closed subprescheme) of X .* I | 121

Proof. Since a locally closed subset of a locally closed subspace of X is a locally closed subspace of X , it is clear (4.1.5) that the question is local and that we can thus suppose that X is affine; the proposition then follows from the canonical identification of $A/\mathfrak{J}$ with $(A/\mathfrak{J})/(\mathfrak{J}'/\mathfrak{J})$, where A is some ring, and $\mathfrak{J}$ and $\mathfrak{J}'$ are ideals of A such that $\mathfrak{J} \subset \mathfrak{J}'$. $\square$

In what follows, we will always make use of the above identification.

(4.1.7). Let Y be a subprescheme of a prescheme X , and denote by ψ the canonical injection $Y \rightarrow X$ of the *underlying subspaces*; we know that the inverse image $\psi^*(\mathcal{O}_X)$ is the restriction $\mathcal{O}_X|_Y$ (0, 3.7.1). If, for each $y \in Y$, we denote by ω_y the canonical homomorphism $(\mathcal{O}_X)_y \rightarrow (\mathcal{O}_Y)_y$, then these homomorphisms are the restrictions to stalks of a *surjective* homomorphism ω of sheaves of rings $\mathcal{O}_X|_Y \rightarrow \mathcal{O}_Y$: indeed, it suffices to check locally on Y , that is to say, we can suppose that X is affine and that the subprescheme Y is closed; if in this case $\mathcal{J}$ is the sheaf of ideals of $\mathcal{O}_X$ which defines Y , then the ω_y are none other than the restrictions to stalks of the homomorphism $\mathcal{O}_X|_Y \rightarrow (\mathcal{O}_X/\mathcal{J})|_Y$. We have thus defined a *monomorphism of ringed spaces* (0, 4.1.1) $j = (\psi, \omega^\flat)$ which is evidently a morphism $Y \rightarrow X$ of preschemes (2.2.1), and we call this the *canonical injection morphism*.

If $f : X \rightarrow Z$ is a morphism, we then say that the composite morphism $Y \xrightarrow{j} X \xrightarrow{f} Z$ is the *restriction of f to the subprescheme Y* .

(4.1.8). Conforming to the general definitions (T, I, 1.1), we will say that a morphism of preschemes $f : Z \rightarrow X$ is *majoré*⁸ by the injection morphism $j : Y \rightarrow X$ of a subprescheme Y of X if f factors as $Z \xrightarrow{g} Y \xrightarrow{j} X$, where g is a morphism of preschemes; further, g is necessarily *unique* since j is a monomorphism.

Proposition (4.1.9). — *For a morphism $f : Z \rightarrow X$ to factor through an injection morphism $j : Y \rightarrow X$, it is necessary and sufficient that $f(Z) \subset Y$ and, for all $z \in Z$, letting $y = f(z)$, that the homomorphism $(\mathcal{O}_X)_y \rightarrow \mathcal{O}_z$ corresponding to f factor as $(\mathcal{O}_X)_y \rightarrow (\mathcal{O}_Y)_y \rightarrow \mathcal{O}_z$ (or equivalently, for the kernel of $(\mathcal{O}_X)_y \rightarrow \mathcal{O}_z$ to contain the kernel of $(\mathcal{O}_X)_y \rightarrow (\mathcal{O}_Y)_y$).*

Proof. The conditions are evidently necessary. To see that they are sufficient, we can reduce to the case where Y is a *closed* subprescheme of X , by replacing X by an open U such that Y is closed in U (4.1.3) if necessary; Y is then defined by a quasi-coherent sheaf of ideals $\mathcal{J}$ of $\mathcal{O}_X$. Let $f = (\psi, \theta)$, and let $\mathcal{J}$ be the sheaf of ideals of $\psi^*(\mathcal{O}_X)$ which is the kernel of $\theta^\sharp : \psi^*(\mathcal{O}_X) \rightarrow \mathcal{O}_Z$; considering the properties of the functor ψ^* (0, 3.7.2), the hypotheses imply that, for each $z \in Z$, we have $(\psi^*(\mathcal{J}))_z \subset \mathcal{J}_z$, and, as a result, that $\psi^*(\mathcal{J}) \subset \mathcal{J}$. Thus $\theta^\sharp$ factors as

$$\psi^*(\mathcal{O}_X) \longrightarrow \psi^*(\mathcal{O}_X)/\psi^*(\mathcal{J}) = \psi^*(\mathcal{O}_X/\mathcal{J}) \xrightarrow{\omega} \mathcal{O}_Z,$$

the first arrow being the canonical homomorphism. Let ψ' be the continuous map $Z \rightarrow Y$ coinciding with ψ ; it is clear that we have $\psi'^*(\mathcal{O}_Y) = \psi^*(\mathcal{O}_X/\mathcal{J})$; on the other hand, ω is evidently a local homomorphism, so $g = (\psi', \omega^\flat)$ is a morphism of preschemes $Z \rightarrow Y$ (2.2.1), which, according to the above, is such that $f = j \circ g$, hence the proposition. $\square$

Corollary (4.1.10). — *For an injection morphism $Z \rightarrow X$ to factor through the injection morphism $Y \rightarrow X$, it is necessary and sufficient for Z to be a subprescheme of Y .*

⁸[Trans.] There doesn't seem to be an English equivalent of this, except for 'bounded above', which doesn't make much sense in this context. We would normally just say that ' f factors through j ', but to avoid having to entirely restructure the often-lengthy sentences in the original, we sometimes (but as little as we can) use 'majoré'.

We then write $Z \leq Y$, and this condition is evidently an *ordering* on the set of subpreschemes of X .

4.2. Immersion morphisms

Definition (4.2.1). — We say that a morphism $f : Y \rightarrow X$ is an immersion (resp. a closed immersion, an open immersion) if it factors as $Y \xrightarrow{g} Z \xrightarrow{j} X$, where g is an isomorphism, Z is a subprescheme of X (resp. a closed subprescheme, a subprescheme induced by an open set), and j is the injection morphism.

The subprescheme Z and the isomorphism g are then determined in a *unique* way, since if Z' is a second subprescheme of X , j' the injection $Z' \rightarrow X$, and g' an isomorphism $Y \rightarrow Z'$ such that $j \circ g = j' \circ g'$, then we have $j' = j \circ g \circ g'^{-1}$, hence $Z' \leq Z$ (4.1.10), and we can similarly show that $Z \leq Z'$, hence $Z' = Z$, and, since j is a monomorphism of preschemes, $g' = g$.

We say that $f = j \circ g$ is the *canonical factorization* of the immersion f , and the subprescheme Z and the isomorphism g are those *associated* to f .

It is clear that an immersion is a *monomorphism* of preschemes (4.1.7) and *a fortiori* a *radicial* morphism (3.5.4).

Proposition (4.2.2). —

- (a) For a morphism $f = (\psi, \theta) : Y \rightarrow X$ to be an open immersion, it is necessary and sufficient for ψ to be a homeomorphism from Y to some open subset of X , and, for all $y \in Y$, that the homomorphism $\theta_y^\# : \mathcal{O}_{\psi(y)} \rightarrow \mathcal{O}_y$ be bijective.
- (b) For a morphism $f = (\psi, \theta) : Y \rightarrow X$ to be an immersion (resp. a closed immersion), it is necessary and sufficient for ψ to be a homeomorphism from Y to some locally closed (resp. closed) subset of X , and, for all $y \in Y$, that the homomorphism $\theta_y^\# : \mathcal{O}_{\psi(y)} \rightarrow \mathcal{O}_y$ be surjective.

Proof.

- (a) The conditions are evidently necessary. Conversely, if they are satisfied, then it is clear that $\theta^\#$ is an isomorphism from $\psi^*(\mathcal{O}_X)$ to $\mathcal{O}_Y$ ⁹, and $\psi^*(\mathcal{O}_X)$ is the sheaf induced by “transport of structure” via ψ^{-1} from $\mathcal{O}_X|_{\psi(Y)}$; hence the conclusion.
- (b) The conditions are evidently necessary—we prove that they are sufficient. Consider first the particular case where we suppose that X is an affine scheme, and that $Z = \psi(Y)$ is *closed* in X . We then know (0, 3.4.6) that $\psi_*(\mathcal{O}_Y)$ has support equal to Z , and that, denoting its restriction to Z by $\mathcal{O}'_Z$, the ringed space $(Z, \mathcal{O}'_Z)$ is induced from $(Y, \mathcal{O}_Y)$ by transport of structure via the homeomorphism ψ considered as a map from Y to Z . Let us now show that $f_*(\mathcal{O}_Y) = \psi_*(\mathcal{O}_Y)$ is a *quasi-coherent* $\mathcal{O}_X$ -module. Indeed, for all $x \notin Z$, $\psi_*(\mathcal{O}_Y)$ restricted to a suitable neighbourhood of x is zero. If instead $x \in Z$, then we have $x = \psi(y)$ for a unique $y \in Y$; let V be an affine open neighbourhood of y in Y ; $\psi(V)$ is then open in Z , and so equal to the intersection of Z with an open subset U of X , and the restriction to U of $\psi_*(\mathcal{O}_Y)$ is identical to the restriction to U of the direct image $(\psi_V)_*(\mathcal{O}_Y|_V)$, where ψ_V is the restriction of ψ to V . The restriction of the morphism (ψ, θ) to $(V, \mathcal{O}_Y|_V)$ is a morphism from this prescheme to $(X, \mathcal{O}_X)$, and, as a result, is of the form $(^a\phi, \tilde{\phi})$, where ϕ is the homomorphism from the ring $A = \Gamma(X, \mathcal{O}_X)$ to the ring $\Gamma(V, \mathcal{O}_Y)$ (1.7.3); we conclude that $(\psi_V)_*(\mathcal{O}_Y|_V)$ is a quasi-coherent $\mathcal{O}_X$ -module (1.6.3), which proves our assertion, due to the local nature of quasi-coherent sheaves. In addition, the hypothesis that ψ is a homeomorphism implies (0, 3.4.5) that, for all $y \in Y$, ψ_y is an isomorphism $(\psi^*(\mathcal{O}_X))_{\psi(y)} \rightarrow \mathcal{O}_y$; since the diagram

$$\begin{array}{ccc} \mathcal{O}_{\psi(y)} & \xrightarrow{\theta_{\psi(y)}} & (\psi^*(\mathcal{O}_X))_{\psi(y)} \\ \psi_y \circ \rho_{\psi(y)} \downarrow & & \downarrow \psi_y \\ (\psi^*(\mathcal{O}_X))_y & \xrightarrow{\theta_y^\#} & \mathcal{O}_y \end{array}$$

⁹[Trans.] The French source reverses the source and target here. By the definition of a morphism of preschemes and by the displayed stalk maps in Proposition 4.2.2(a), $\theta^\#$ has source $\psi^*(\mathcal{O}_X)$ and target $\mathcal{O}_Y$.

is commutative, and the vertical arrows are the isomorphisms (0, 3.7.2), the hypothesis that $\theta_y^\sharp$ is surjective implies that $\theta_{\psi(y)}$ is surjective as well. Since the support of $\psi_*(\mathcal{O}_Y)$ is $Z = \psi(Y)$, θ is a *surjective* homomorphism from $\mathcal{O}_X = \tilde{A}$ to the quasi-coherent $\mathcal{O}_X$ -module $f_*(\mathcal{O}_Y)$. As a result, there exists a unique isomorphism ω from a sheaf quotient $\tilde{A}/\tilde{\mathfrak{J}}$ (with $\mathfrak{J}$ being an ideal of A) to $f_*(\mathcal{O}_Y)$ which, when composed with the canonical homomorphism $\tilde{A} \rightarrow \tilde{A}/\tilde{\mathfrak{J}}$, gives θ (1.3.8); if $\mathcal{O}_Z$ denotes the restriction of $\tilde{A}/\tilde{\mathfrak{J}}$ to Z , then $(Z, \mathcal{O}_Z)$ is a subprescheme of $(X, \mathcal{O}_X)$, and f factors through the canonical injection of this subprescheme into X and the isomorphism (ψ_0, ω_0) , where ψ_0 is ψ considered as a map from Y to Z , and ω_0 the restriction of ω to $\mathcal{O}_Z$.

We now pass to the general case. Let U be an affine open subset of X such that $U \cap \psi(Y)$ is closed in U and nonempty. By restricting f to the prescheme induced by Y on the open subset $\psi^{-1}(U)$, and by considering it as a morphism from this prescheme to the prescheme induced by X on U , we reduce to the first case; the restriction of f to $\psi^{-1}(U)$ is thus a closed immersion $\psi^{-1}(U) \rightarrow U$, canonically factoring as $j_U \circ g_U$, where g_U is an isomorphism from the prescheme $\psi^{-1}(U)$ to a subprescheme Z_U of U , and j_U is the canonical injection $Z_U \rightarrow U$. Let V be a second affine open subset of X such that $V \subset U$; since the restriction Z'_V of Z_U to V is a subprescheme of the prescheme V , the restriction of f to $\psi^{-1}(V)$ factors as $j'_V \circ g'_V$, where j'_V is the canonical injection $Z'_V \rightarrow V$ and g'_V is an isomorphism from $\psi^{-1}(V)$ to Z'_V . By the uniqueness of the canonical factorization of an immersion (4.2.1), we necessarily have that $Z'_V = Z_V$ and $g'_V = g_V$. We conclude (4.1.5) that there is a subprescheme Z of X whose underlying space is $\psi(Y)$, and whose restriction to each $U \cap \psi(Y)$ is Z_U ; the g_U are then the restrictions to $\psi^{-1}(U)$ of an isomorphism $g : Y \rightarrow Z$ such that $f = j \circ g$, where j is the canonical injection $Z \rightarrow X$.

□

Corollary (4.2.3). — *Let X be an affine scheme. For a morphism $f = (\psi, \theta) : Y \rightarrow X$ to be a closed immersion, it is necessary and sufficient for Y to be an affine scheme, and the homomorphism $\Gamma(\psi) : \Gamma(\mathcal{O}_X) \rightarrow \Gamma(\mathcal{O}_Y)$ to be surjective.*

Corollary (4.2.4). —

- (a) *Let f be a morphism $Y \rightarrow X$, and (V_λ) a cover of $f(Y)$ by open subsets of X . For f to be an immersion (resp. an open immersion), it is necessary and sufficient for its restriction to each of the induced preschemes $f^{-1}(V_\lambda)$ to be an immersion (resp. an open immersion) into V_λ .*
- (b) *Let f be a morphism $Y \rightarrow X$, and (V_λ) an open cover of X . For f to be a closed immersion, it is necessary and sufficient for its restriction to each of the induced preschemes $f^{-1}(V_\lambda)$ to be a closed immersion into V_λ .*

I | 124

Proof. Let $f = (\psi, \theta)$; in the case (a), $\theta_y^\sharp$ is surjective (resp. bijective) for all $y \in Y$, and in the case (b), $\theta_y^\sharp$ is surjective for all $y \in Y$; it thus suffices to check, in case (a), that ψ is a homeomorphism from Y to a locally closed (resp. open) subset of X , and, in case (b), that ψ is a homeomorphism from Y to a closed subset of X . Now ψ is evidently injective, and sends each neighbourhood of y in Y to a neighbourhood of $\psi(y)$ in $\psi(Y)$ for all $y \in Y$, by virtue of the hypothesis; in case (a), $\psi(Y) \cap V_\lambda$ is locally closed (resp. open) in V_λ , so $\psi(Y)$ is locally closed (resp. open) in the union of the V_λ , and *a fortiori* in X ; in case (b), $\psi(Y) \cap V_\lambda$ is closed in V_λ , so $\psi(Y)$ is closed in X since $X = \bigcup_\lambda V_\lambda$. □

Proposition (4.2.5). — *The composition of any two immersions (resp. of two open immersions, of two closed immersions) is an immersion (resp. an open immersion, a closed immersion).*

Proof. This follows easily from (4.1.6). □

4.3. Products of immersions

Proposition (4.3.1). — Let $\alpha : X' \rightarrow X$, $\beta : Y' \rightarrow Y$ be two S -morphisms; if α and β are immersions (resp. open immersions, closed immersions), then $\alpha \times_S \beta$ is an immersion (resp. an open immersion, a closed immersion). In addition, if α (resp. β) identifies X' (resp. Y') with a subprescheme X'' (resp. Y'') of X (resp. Y), then $\alpha \times_S \beta$ identifies the underlying space of $X' \times_S Y'$ with the subspace $p^{-1}(X'') \cap q^{-1}(Y'')$ of the underlying space of $X \times_S Y$, where p and q denote the projections from $X \times_S Y$ to X and Y respectively.

Proof. According to Definition (4.2.1), we can restrict to the case where X' and Y' are subpreschemes, and α and β are the injection morphisms. The proposition has already been proved for the subpreschemes induced by open sets (3.2.7); since every subprescheme is a closed subprescheme of a prescheme induced by an open set (4.1.3), we can reduce to the case where X' and Y' are closed subpreschemes.

Let us first show that we can assume S is *affine*. Let (S_λ) be a cover of S by affine open sets; if ϕ and ψ are the structure morphisms of X and Y , then let $X_\lambda = \phi^{-1}(S_\lambda)$ and $Y_\lambda = \psi^{-1}(S_\lambda)$. The restriction X'_λ (resp. Y'_λ) of X' (resp. Y') to $X_\lambda \cap X'$ (resp. $Y_\lambda \cap Y'$) is a closed subprescheme of X_λ (resp. Y_λ), the preschemes $X_\lambda, Y_\lambda, X'_\lambda$, and Y'_λ can be considered as S_λ -preschemes, and the products $X_\lambda \times_S Y_\lambda$ and $X_\lambda \times_{S_\lambda} Y_\lambda$ (resp. $X'_\lambda \times_S Y'_\lambda$ and $X'_\lambda \times_{S_\lambda} Y'_\lambda$) are identical (3.2.5). If the proposition is true when S is affine, then the restriction of $\alpha \times_S \beta$ to each of the $X'_\lambda \times_S Y'_\lambda$ is thus an immersion (3.2.7). Since the product $X'_\lambda \times_S Y'_\mu$ (resp. $X_\lambda \times_S Y_\mu$) can be identified with $(X'_\lambda \cap X'_\mu) \times_S (Y'_\lambda \cap Y'_\mu)$ (resp. $(X_\lambda \cap X_\mu) \times_S (Y_\lambda \cap Y_\mu)$) (3.2.6.4), the restriction of $\alpha \times_S \beta$ to each of the $X'_\lambda \times_S Y'_\mu$ is also an immersion; the same is true for $\alpha \times_S \beta$ by (4.2.4). I | 125

Next, we show that we can assume X and Y are *affine*. Indeed, let (U_i) (resp. (V_j)) be a cover of X (resp. Y) by affine open sets, and let X'_i (resp. Y'_j) be the restriction of X' (resp. Y') to $X' \cap U_i$ (resp. $Y' \cap V_j$), which is a closed subprescheme of U_i (resp. V_j); $U_i \times_S V_j$ can be identified with the restriction of $X \times_S Y$ to $p^{-1}(U_i) \cap q^{-1}(V_j)$ (3.2.7); similarly, if p' and q' are the projections from $X' \times_S Y'$, then $X'_i \times_S Y'_j$ can be identified with the restriction of $X' \times_S Y'$ to $p'^{-1}(X'_i) \cap q'^{-1}(Y'_j)$. Set $\gamma = \alpha \times_S \beta$; we have, by definition, $p \circ \gamma = \alpha \circ p'$ and $q \circ \gamma = \beta \circ q'$; since $X'_i = \alpha^{-1}(U_i)$ and $Y'_j = \beta^{-1}(V_j)$, we also have $p'^{-1}(X'_i) = \gamma^{-1}(p^{-1}(U_i))$ and $q'^{-1}(Y'_j) = \gamma^{-1}(q^{-1}(V_j))$, hence

$$p'^{-1}(X'_i) \cap q'^{-1}(Y'_j) = \gamma^{-1}(p^{-1}(U_i) \cap q^{-1}(V_j)) = \gamma^{-1}(U_i \times_S V_j),$$

and we then conclude as in the previous part of the proof.

So suppose X, Y , and S are affine, and let B, C , and A be their respective rings. Then B and C are A -algebras, and X' and Y' are affine schemes whose rings are quotient algebras B' and C' of B and C respectively. In addition, we have $\alpha = ({}^a\rho, \tilde{\rho})$ and $\beta = ({}^a\sigma, \tilde{\sigma})$, where ρ and σ are (respectively) the canonical homomorphisms $B \rightarrow B'$ and $C \rightarrow C'$ (1.7.3). With that in mind, we know that $X \times_S Y$ (resp. $X' \times_S Y'$) is an affine scheme with ring $B \otimes_A C$ (resp. $B' \otimes_A C'$), and $\alpha \times_S \beta = ({}^a\tau, \tilde{\tau})$, where τ is the homomorphism $\rho \otimes \sigma$ from $B \otimes_A C$ to $B' \otimes_A C'$ (Proposition (3.2.2) and Corollary (3.2.3)); since this homomorphism is surjective, $\alpha \times_S \beta$ is an immersion. In addition, if $\mathfrak{b}$ (resp. $\mathfrak{c}$) is the kernel of ρ (resp. σ), then the kernel of τ is $u(\mathfrak{b}) + v(\mathfrak{c})$, where u (resp. v) is the homomorphism $b \mapsto b \otimes 1$ (resp. $c \mapsto 1 \otimes c$). Since $p = ({}^au, \tilde{u})$ and $q = ({}^av, \tilde{v})$, this kernel corresponds, in the prime spectrum of $B \otimes_A C$, to the closed set $p^{-1}(X') \cap q^{-1}(Y')$, by (1.2.2.1) and Proposition (1.1.2, iii), which finishes the proof. $\square$

Corollary (4.3.2). — If $f : X \rightarrow Y$ is an immersion (resp. an open immersion, a closed immersion) and an S -morphism, then $f_{(S')}$ is an immersion (resp. an open immersion, a closed immersion) for every extension $S' \rightarrow S$ of the base prescheme.

4.4. Inverse images of a subprescheme

Proposition (4.4.1). — Let $f : X \rightarrow Y$ be a morphism, Y' a subprescheme (resp. a closed subprescheme, a prescheme induced by an open set) of Y , and $j : Y' \rightarrow Y$ the injection morphism. Then the projection $p : X \times_Y Y' \rightarrow X$ is an immersion (resp. a closed immersion, an open immersion); the underlying space of the subprescheme of X associated to p is $f^{-1}(Y')$; in addition, if j' is the injection morphism of this subprescheme into X , then, for every morphism $h : Z \rightarrow X$, the composite $f \circ h : Z \rightarrow Y$ factors through j if and only if h factors through j' .

Proof. Since $p = 1_X \times_Y j$ (3.3.4), the first claim follows from Proposition (4.3.1); the second is a particular case of Corollary (3.5.10) (after swapping the roles of X and Y'). Finally, if we have $f \circ h = j \circ h'$, where h' is a morphism $Z \rightarrow Y'$, then it follows from the definition of the product that we have $h = p \circ u$, where u is a morphism $Z \rightarrow X \times_Y Y'$, whence the last claim. $\square$

We say that the subprescheme of X thus defined is the *inverse image* of the subprescheme Y' of Y under the morphism f , terminology which is consistent with that introduced more generally in (3.3.6). When we speak of $f^{-1}(Y')$ as a subprescheme of X , this will always be the subprescheme we mean.

When the preschemes $f^{-1}(Y')$ and X are identical, j' is the identity and each morphism $h : Z \rightarrow X$ thus factors through j' ; taking $h = 1_X$, we see that the morphism $f : X \rightarrow Y$ factors as $X \xrightarrow{g} Y' \xrightarrow{j} Y$.

When y is a closed point of Y and $Y' = \text{Spec}(k(y))$ is the smallest closed subprescheme of Y having $\{y\}$ as its underlying space (4.1.9), the closed subprescheme $f^{-1}(Y')$ is canonically isomorphic to the fibre $f^{-1}(y)$ defined in (3.6.2), and we will use this identification in all that follows.

Corollary (4.4.2). — Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be morphisms, and $h = g \circ f$ their composition. For each subprescheme Z' of Z , the subpreschemes $f^{-1}(g^{-1}(Z'))$ and $h^{-1}(Z')$ of X are identical.

Proof. This follows from the existence of the canonical isomorphism $X \times_Y (Y \times_Z Z') \simeq X \times_Z Z'$ (3.3.9.1). $\square$

Corollary (4.4.3). — Let X' and X'' be subpreschemes of X , and let $j' : X' \rightarrow X$ and $j'' : X'' \rightarrow X$ be their injection morphisms; then $j'^{-1}(X'')$ and $j''^{-1}(X')$ are both equal to the greatest lower bound $\inf(X', X'')$ of X' and X'' for the ordering $\leq$ on subpreschemes, and this is canonically isomorphic to $X' \times_X X''$.

Proof. This follows immediately from Proposition (4.4.1) and Corollary (4.1.10). $\square$

Corollary (4.4.4). — Let $f : X \rightarrow Y$ be a morphism, and Y' and Y'' subpreschemes of Y ; then we have $f^{-1}(\inf(Y', Y'')) = \inf(f^{-1}(Y'), f^{-1}(Y''))$.

Proof. This follows from the existence of the canonical isomorphism between $(X \times_Y Y') \times_X (X \times_Y Y'')$ and $X \times_Y (Y' \times_Y Y'')$ (3.3.9.1). $\square$

Proposition (4.4.5). — Let $f : X \rightarrow Y$ be a morphism, and Y' a closed subprescheme of Y defined by a quasi-coherent sheaf of ideals $\mathcal{K}$ of $\mathcal{O}_Y$ (4.1.3); the closed subprescheme $f^{-1}(Y')$ of X is then defined by the quasi-coherent sheaf of ideals $f^*(\mathcal{K})\mathcal{O}_X$ of $\mathcal{O}_X$.

Proof. The statement is evidently local on X and Y ; it thus suffices to note that if A is a B -algebra and $\mathfrak{K}$ an ideal of B , then we have $A \otimes_B (B/\mathfrak{K}) = A/\mathfrak{K}A$, and then to apply (1.6.9). $\square$

Corollary (4.4.6). — Let X' be a closed subprescheme of X defined by a quasi-coherent sheaf of ideals $\mathcal{J}$ of $\mathcal{O}_X$, and i the injection $X' \rightarrow X$; for the restriction $f \circ i$ of f to X' to factor through the injection $j : Y' \rightarrow Y$ (in other words, for it to factor as $j \circ g$, with g a morphism $X' \rightarrow Y'$), it is necessary and sufficient that $f^*(\mathcal{K})\mathcal{O}_X \subset \mathcal{J}$.

Proof. It suffices to apply Proposition (4.4.1) to i , taking Proposition (4.4.5) into account. $\square$

4.5. Local immersions and local isomorphisms

Definition (4.5.1). — Let $f : X \rightarrow Y$ be a morphism of preschemes. We say that f is a local immersion at a point $x \in X$ if there exists an open neighbourhood U of x in X and an open neighbourhood V of $f(x)$ in Y such that the restriction of f to the induced prescheme U is a closed immersion of U into the induced prescheme V . We say that f is a local immersion if f is a local immersion at each point of X .

Definition (4.5.2). — We say that a morphism $f : X \rightarrow Y$ is a local isomorphism at a point $x \in X$ if there exists an open neighbourhood U of x in X such that the restriction of f to the induced prescheme U is an open immersion of U into Y . We say that f is a local isomorphism if f is a local isomorphism at each point of X . I | 127

(4.5.3). An immersion (resp. a closed immersion) $f : X \rightarrow Y$ can be characterized as a local immersion such that f is a homeomorphism from the underlying space of X to a subset (resp. a closed subset) of Y . An open immersion f can be characterized as an *injective* local isomorphism.

Proposition (4.5.4). — Let X be an irreducible prescheme, and $f : X \rightarrow Y$ a dominant injective morphism. If f is a local immersion, then f is an immersion, and $f(X)$ is open in Y .

Proof. Let $x \in X$, and choose an open neighbourhood U of x and an open neighbourhood V of $f(x)$ in Y such that the restriction of f to U is a closed immersion into V ; since U is dense in X , $f(U)$ is dense in V by hypothesis, so $f(U) = V$, and f is a homeomorphism from U to V ; the hypothesis that f is injective implies that $f^{-1}(V) = U$, hence the proposition. $\square$

Proposition (4.5.5). —

- (i) The composition of any two local immersions (resp. of two local isomorphisms) is a local immersion (resp. a local isomorphism).
- (ii) Let $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ be two S -morphisms. If f and g are local immersions (resp. local isomorphisms), then so too is $f \times_S g$.
- (iii) If an S -morphism f is a local immersion (resp. a local isomorphism), then so too is $f_{(S')}$ for every extension $S' \rightarrow S$ of the base prescheme.

Proof. According to (3.5.1), it suffices to prove (i) and (ii).

(i) follows immediately from the transitivity of closed (resp. open) immersions (4.2.5)¹⁰ and from the fact that if f is a homeomorphism from X to a closed subset of Y , then for every open $U \subset X$, $f(U)$ is open in $f(X)$, so there exists an open subset V of Y such that $f(U) = V \cap f(X)$, and, as a result, $f(U)$ is closed in V .

To prove (ii), let $z \in X \times_S Y$, set $z' = (f \times_S g)(z)$ ¹¹, let p and q be the projections from $X \times_S Y$, and let p' and q' be the projections from $X' \times_S Y'$. There exist, by hypothesis, open neighbourhoods U, U', V , and V' of $x = p(z)$, $x' = p'(z')$, $y = q(z)$, and $y' = q'(z')$ (respectively), such that the restrictions of f and g to U and V (respectively) are closed (resp. open) immersions into U' and V' (respectively). Since the underlying spaces of $U \times_S V$ and $U' \times_S V'$ can be identified with the open neighbourhoods $p^{-1}(U) \cap q^{-1}(V)$ and $p'^{-1}(U') \cap q'^{-1}(V')$ of z and z' (respectively) (3.2.7), the proposition follows from Proposition (4.3.1). $\square$

§5. REDUCED PRESCHMES; THE SEPARATION CONDITION

5.1. Reduced preschemes

Proposition (5.1.1). — Let $(X, \mathcal{O}_X)$ be a prescheme, and $\mathcal{B}$ a quasi-coherent $\mathcal{O}_X$ -algebra. Then there exists a unique quasi-coherent $\mathcal{O}_X$ -module $\mathcal{N}$ whose stalk $\mathcal{N}_x$ at any $x \in X$ is the nilradical of the ring $\mathcal{B}_x$. When X is affine, and, consequently, $\mathcal{B} = \tilde{B}$, where B is an algebra over $A(X)$, then we have $\mathcal{N} = \tilde{\mathfrak{N}}$, where $\mathfrak{N}$ is the nilradical of B .

¹⁰[Trans.] The French source cites (4.2.4) here; transitivity is Proposition 4.2.5.

¹¹[Trans.] The French source uses z and z' below without introducing them; this insertion supplies the required point and its image.

Proof. The statement is local, so it suffices to show the latter claim. We know that $\tilde{\mathfrak{N}}$ is a quasi-coherent $\mathcal{O}_X$ -module (1.4.1), and that its stalk at a point $x \in X$ is the ideal $\mathfrak{N}_x$ of the ring of fractions B_x ; it remains to prove that the nilradical of B_x is contained in $\mathfrak{N}_x$, the converse inclusion being evident. Let z/s be an element of the nilradical of B_x , with $z \in B$, and $s \notin \mathfrak{j}_x$; by hypothesis, there exists an integer k such that $(z/s)^k = 0$, which implies that there exists some $t \notin \mathfrak{j}_x$ such that $tz^k = 0$. We conclude that $(tz)^k = 0$, and, as a result, that $z/s = (tz)/(ts) \in \mathfrak{N}_x$. $\square$

We say that the quasi-coherent $\mathcal{O}_X$ -module $\mathcal{N}$ thus defined is the *nilradical* of the $\mathcal{O}_X$ -algebra $\mathcal{B}$; in particular, we denote by $\mathcal{N}_X$ the nilradical of $\mathcal{O}_X$.

Corollary (5.1.2). — *Let X be a prescheme; the closed subscheme of X defined by the sheaf of ideals $\mathcal{N}_X$ is the only reduced subscheme (0, 4.1.4) of X that has X as its underlying space; it is also the smallest subscheme of X that has X as its underlying space.*

Proof. Let Y denote the closed subscheme of X defined by $\mathcal{N}_X$. Since its structure sheaf is $\mathcal{O}_X/\mathcal{N}_X$, it is immediate that Y is reduced and has X as its underlying space, because $\mathcal{N}_x \neq \mathcal{O}_x$ for every $x \in X$. To show the other claims, note that a subscheme Z of X that has X as its underlying space is defined by a sheaf of ideals $\mathcal{I}$ (4.1.3) such that $\mathcal{I}_x \neq \mathcal{O}_x$ for any $x \in X$. We can restrict to the case where X is affine, say $X = \text{Spec}(A)$ and $\mathcal{I} = \tilde{\mathfrak{I}}$, where $\mathfrak{I}$ is an ideal of A ; then, for every $x \in X$, we have $\mathfrak{I} \subset \mathfrak{j}_x$, and so $\mathfrak{I}$ is contained in every prime ideal of A , and so also in their intersection $\mathfrak{N}$, the nilradical of A . This proves that Y is the smallest subscheme of X that has X as its underlying space (4.1.9); furthermore, if Z is distinct from Y , we necessarily have $\mathcal{I}_x \neq \mathcal{N}_x$ for at least one $x \in X$, and so (5.1.1) Z is not reduced. $\square$

Definition (5.1.3). — We define the reduced prescheme associated to a prescheme X , denoted by X_{red} , to be the unique reduced subscheme of X that has X as its underlying space.

Saying that a prescheme X is reduced thus means that $X = X_{\text{red}}$.

Proposition (5.1.4). — *For the prime spectrum of a ring A to be a reduced (resp. integral) prescheme (2.1.7), it is necessary and sufficient for A to be a reduced (resp. integral) ring.*

Proof. Indeed, it follows immediately from (5.1.1) that the condition $\mathcal{N} = (0)$ is necessary and sufficient for $X = \text{Spec}(A)$ to be reduced; the claim corresponding to integral rings is then a consequence of (1.1.13). $\square$

Since every nonzero ring of fractions of an integral ring is integral, it follows from (5.1.4) that, for every *locally integral* prescheme X , $\mathcal{O}_x$ is an *integral* ring for every $x \in X$. The converse is true whenever the underlying space of X is *locally Noetherian*: indeed, X is then reduced, and if U is an affine open subset of X , which is a Noetherian space, then U has only a finite number of irreducible components, and so its ring A has only a finite number of minimal prime ideals (1.1.14). If two of the irreducible components of U had a common point x , then $\mathcal{O}_x$ would have at least two distinct minimal prime ideals, and would thus not be integral; the components of U are thus open subsets that are pairwise disjoint, and each of them is thus integral.

(5.1.5). Let $f = (\psi, \theta) : X \rightarrow Y$ be a morphism of preschemes; the homomorphism $\theta_x^\# : \mathcal{O}_{\psi(x)} \rightarrow \mathcal{O}_x$ sends each nilpotent element of $\mathcal{O}_{\psi(x)}$ to a nilpotent element of $\mathcal{O}_x$; by passing to the quotients, $\theta^\#$ induces a homomorphism

$$\omega : \psi^*(\mathcal{O}_Y/\mathcal{N}_Y) \longrightarrow \mathcal{O}_X/\mathcal{N}_X.$$

It is clear that, for every $x \in X$, $\omega_x : \mathcal{O}_{\psi(x)}/\mathcal{N}_{\psi(x)} \rightarrow \mathcal{O}_x/\mathcal{N}_x$ is a local homomorphism, and so $(\psi, \omega^\flat)$ is a morphism of preschemes $X_{\text{red}} \rightarrow Y_{\text{red}}$, which we denote by f_{red} , and call the *reduced* morphism associated to f . It is immediate that, for morphisms $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, we have $(g \circ f)_{\text{red}} = g_{\text{red}} \circ f_{\text{red}}$, and so we have defined X_{red} as a *functor, covariant* in X .

The preceding definition shows that the diagram

$$\begin{array}{ccc} X_{\text{red}} & \xrightarrow{f_{\text{red}}} & Y_{\text{red}} \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

is commutative, where the vertical arrows are the injection morphisms; in other words, $X_{\text{red}} \rightarrow X$ is a *functorial* morphism. We note in particular that, if X is reduced, then every morphism $f : X \rightarrow Y$ factors as $X \xrightarrow{f_{\text{red}}} Y_{\text{red}} \rightarrow Y$; in other words, f factors through the injection morphism $Y_{\text{red}} \rightarrow Y$.

Proposition (5.1.6). — *Let $f : X \rightarrow Y$ be a morphism; if f is surjective (resp. radicial, an immersion, a closed immersion, an open immersion, a local immersion, a local isomorphism), then so too is f_{red} . Conversely, if f_{red} is surjective (resp. radicial), then so too is f .*

Proof. The proposition is trivial if f is surjective; if f is radicial, then the proposition follows from the fact that, for every $x \in X$, the field $k(x)$ is the same for the preschemes X and X_{red} (3.5.8). Finally, if $f = (\psi, \theta)$ is an immersion, a closed immersion, or a local immersion (resp. an open immersion, or a local isomorphism), then the proposition follows from the fact that, if $\theta_x^\#$ is surjective (resp. bijective), then so too is the homomorphism obtained by passing to the quotients by the nilradicals $\mathcal{O}_{\psi(x)}$ and $\mathcal{O}_x$ ((5.1.2) and (4.2.2)) (cf. (5.5.12)). $\square$

Proposition (5.1.7). — *If X and Y are S -preschemes, then the preschemes $X_{\text{red}} \times_{S_{\text{red}}} Y_{\text{red}}$ and $X_{\text{red}} \times_S Y_{\text{red}}$ are identical, and canonically identified with a subprescheme of $X \times_S Y$ that has the same underlying space as that product.*

Proof. The canonical identification of $X_{\text{red}} \times_S Y_{\text{red}}$ with a subprescheme of $X \times_S Y$ that has the same underlying space follows from (4.3.1). Furthermore, if ϕ and ψ are the structure morphisms $X_{\text{red}} \rightarrow S$ and $Y_{\text{red}} \rightarrow S$ (respectively), then they factor through S_{red} (5.1.5), and since $S_{\text{red}} \rightarrow S$ is a monomorphism, the first claim of the proposition follows from (3.2.4). $\square$

Corollary (5.1.8). — *The preschemes $(X \times_S Y)_{\text{red}}$ and $(X_{\text{red}} \times_{S_{\text{red}}} Y_{\text{red}})_{\text{red}}$ are canonically identified with one another.*

Proof. This follows from (5.1.2) and (5.1.7). $\square$

We note that, even if X and Y are reduced preschemes, $X \times_S Y$ might not be reduced, because the tensor product of two reduced algebras can have nilpotent elements.

Proposition (5.1.9). — *Let X be a prescheme, and $\mathcal{I}$ a quasi-coherent sheaf of ideals of $\mathcal{O}_X$ such that $\mathcal{I}^n = 0$ for some integer $n > 0$. Let X_0 be the closed subprescheme $(X, \mathcal{O}_X / \mathcal{I})$ of X ; for X to be an affine scheme, it is necessary and sufficient for X_0 to be an affine scheme.* I | 130

The condition is clearly necessary, so we will show that it is sufficient. If we set $X_k = (X, \mathcal{O}_X / \mathcal{I}^{k+1})$, it is enough to prove by induction on k that X_k is affine, and so we are led to consider the base case, where $\mathcal{I}^2 = 0$. We set

$$\begin{aligned} A &= \Gamma(X, \mathcal{O}_X) \\ A_0 &= \Gamma(X_0, \mathcal{O}_{X_0}) = \Gamma(X, \mathcal{O}_X / \mathcal{I}). \end{aligned}$$

The canonical homomorphism $\mathcal{O}_X \rightarrow \mathcal{O}_X / \mathcal{I}$ induces a homomorphism of rings $\phi : A \rightarrow A_0$. We will see below that ϕ is *surjective*, which implies that

$$(5.1.9.1) \quad 0 \longrightarrow \Gamma(X, \mathcal{I}) \longrightarrow \Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(X, \mathcal{O}_X / \mathcal{I}) \longrightarrow 0$$

is an *exact* sequence. We now prove, assuming that this is true, the proposition. Note that $\mathfrak{K} = \Gamma(X, \mathcal{I})$ is an ideal whose square is zero in A , and thus a module over $A_0 = A / \mathfrak{K}$. By hypothesis, we have $X_0 = \text{Spec}(A_0)$, and, since the underlying spaces of X_0 and X are identical, $\mathfrak{K} = \Gamma(X_0, \mathcal{I})$; Additionally, since $\mathcal{I}^2 = 0$, $\mathcal{I}$ is a quasi-coherent $(\mathcal{O}_X / \mathcal{I})$ -module, so we have $\mathcal{I} \cong \tilde{\mathfrak{K}}$ and $\mathfrak{K}_x = \mathcal{I}_x$ for all $x \in X_0$ (1.4.1). With this in mind, let $X' = \text{Spec}(A)$, and consider the morphism $f = (\psi, \theta) : X \rightarrow X'$ of preschemes that corresponds to the identity map $A \rightarrow \Gamma(X, \mathcal{O}_X)$ (2.2.4). For every affine open subset V of X , the diagram

$$\begin{array}{ccc} A & \longrightarrow & \Gamma(V, \mathcal{O}_X|V) \\ \downarrow & & \downarrow \\ A_0 = A / \mathfrak{K} & \longrightarrow & \Gamma(V, \mathcal{O}_{X_0}|V) \end{array}$$

commutes, whence the diagram

$$\begin{array}{ccc} X' & \xleftarrow{f} & X \\ j' \uparrow & & \uparrow j \\ X'_0 & \xleftarrow{f_0} & X_0 \end{array}$$

also commutes (X'_0 being the closed subprescheme of X' defined by the quasi-coherent sheaf of ideals $\tilde{\mathfrak{K}}$, and j and j' being the canonical injection morphisms). But since X_0 is affine, f_0 is an isomorphism, and since the underlying continuous maps of j and j' are identity maps, we see straight away that $\psi : X \rightarrow X'$ is a homeomorphism. Furthermore, the equation $\mathfrak{K}_x = \mathcal{I}_x$ shows that the restriction of $\theta^\# : \psi^*(\mathcal{O}_{X'}) \rightarrow \mathcal{O}_X$ is an isomorphism from $\psi^*(\tilde{\mathfrak{K}})$ to $\mathcal{I}$; additionally, by passing to the quotients, $\theta^\#$ gives an isomorphism $\psi^*(\mathcal{O}_{X'}/\tilde{\mathfrak{K}}) \rightarrow \mathcal{O}_X/\mathcal{I}$, because f_0 is an isomorphism; we thus immediately conclude, by the 5 lemma (M, I, 1.1), that $\theta^\#$ is itself an isomorphism, and thus that f is an isomorphism, and thus that X is affine. So everything reduces to proving the exactness of (5.1.9.1), which will follow from showing that $H^1(X, \mathcal{I}) = 0$. But $H^1(X, \mathcal{I}) = H^1(X_0, \mathcal{I})$, and we have seen that $\mathcal{I}$ is a quasi-coherent $\mathcal{O}_{X_0}$ -module. Our proof will thus follow from

I | 131

Lemma (5.1.9.2). — *If Y is an affine scheme, and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_Y$ -module, then $H^1(Y, \mathcal{F}) = 0$.*

Proof. This lemma will be proven in Chapter III, §1, as a consequence of the more general theorem that $H^i(Y, \mathcal{F}) = 0$ for all $i > 0$. To give an independent proof, note that $H^1(Y, \mathcal{F})$ can be identified with the module $\text{Ext}_{\mathcal{O}_Y}^1(Y; \mathcal{O}_Y, \mathcal{F})$ of extension classes of the $\mathcal{O}_Y$ -module $\mathcal{O}_Y$ by the $\mathcal{O}_Y$ -module $\mathcal{F}$ (T, 4.2.3); so everything reduces to proving that such an extension $\mathcal{G}$ is trivial. But, for every $y \in Y$, there is a neighbourhood V of y in Y such that $\mathcal{G}|_V$ is isomorphic to $\mathcal{F}|_V \oplus \mathcal{O}_Y|_V$ ¹² (0, 5.4.9); from this we conclude that $\mathcal{G}$ is a quasi-coherent $\mathcal{O}_Y$ -module. If A is the ring of Y , then we have $\mathcal{F} = \tilde{M}$ and $\mathcal{G} = \tilde{N}$, where M and N are A -modules, and, by hypothesis, N is an extension of the A -module A by the A -module M (1.3.11). Since this extension is necessarily trivial, the lemma is proven, and thus so too is (5.1.9). $\square$

Corollary (5.1.10). — *Let X be a prescheme such that $\mathcal{N}_X$ is nilpotent. For X to be an affine scheme, it is necessary and sufficient for X_{red} to be an affine scheme.*

5.2. Existence of a subprescheme with a given underlying space

Proposition (5.2.1). — *For every locally closed subspace Y of the underlying space of a prescheme X , there exists exactly one reduced subprescheme of X that has Y as its underlying space.*

Proof. The uniqueness follows from (5.1.2), so it remains only to show the existence of the prescheme in question.

If X is affine with ring A , and Y is closed in X , then the proposition is immediate: $j(Y)$ is the largest ideal $\mathfrak{a} \subset A$ such that $V(\mathfrak{a}) = Y$, and it is equal to its radical (1.1.4, i), so $A/j(Y)$ is a reduced ring.

In the general case, for every affine open $U \subset X$ such that $U \cap Y$ is closed in U , consider the closed subprescheme Y_U of U defined by the sheaf of ideals associated to the ideal $j(U \cap Y)$ of $A(U)$. This subprescheme is reduced. We claim that, if V is an affine open subset of X contained in U , then Y_V is induced by Y_U on $V \cap Y$. Indeed, this induced prescheme is a closed reduced subprescheme of V having $V \cap Y$ as its underlying space; the uniqueness of Y_V therefore proves the claim. $\square$

Proposition (5.2.2). — *Let X be reduced, let $f : X \rightarrow Y$ be a morphism, and let Z be a closed subprescheme of Y such that $f(X) \subset Z$. Then f factors as $X \xrightarrow{g} Z \xrightarrow{j} Y$, where j is the injection morphism.*

Proof. It follows from the hypotheses that the closed subprescheme $f^{-1}(Z)$ of X has all of X as its underlying space (4.4.1); since X is reduced, this closed subprescheme agrees with X (5.1.2), and the proposition then follows from (4.4.1). $\square$

¹²[Trans.] The French source prints $\mathcal{F}|_V$ here; the restriction required on the neighbourhood V is $\mathcal{F}|_V$.

Corollary (5.2.3). — Let X be a reduced subprescheme of a prescheme Y ; if Z is the closed reduced subprescheme of Y that has $\overline{X}$ as its underlying space, then X is a subprescheme induced on an open subset of Z .

Proof. There is indeed an open subset U of Y such that $X = U \cap \overline{X}$; since, by (5.2.2), X is a reduced subprescheme of Z , the subprescheme X is induced by Z on the open subspace X by uniqueness (5.2.1). $\square$

Corollary (5.2.4). — Let $f : X \rightarrow Y$ be a morphism, and X' (resp. Y') a closed subprescheme of X (resp. Y) defined by a quasi-coherent sheaf of ideals $\mathcal{J}$ (resp. $\mathcal{K}$) of $\mathcal{O}_X$ (resp. $\mathcal{O}_Y$). Suppose that X' is reduced, and that $f(X') \subset Y'$. Then $f^*(\mathcal{K})\mathcal{O}_X \subset \mathcal{J}$.

Proof. Since, by (5.2.2), the restriction of f to X' factors as $X' \rightarrow Y' \rightarrow Y$, it suffices to apply (4.4.6). $\square$

5.3. Diagonal; graph of a morphism

(5.3.1). Let X be an S -prescheme; we define the *diagonal morphism* from X into $X \times_S X$, denoted by $\Delta_{X|S}$, or Δ_X , or even Δ if no confusion may arise, to be the S -morphism $(1_X, 1_X)_S$, or, in other words, the unique S -morphism Δ_X such that

$$(5.3.1.1) \quad p_1 \circ \Delta_X = p_2 \circ \Delta_X = 1_X,$$

where p_1 and p_2 are the projections of $X \times_S X$ (Definition (3.2.1)). If $f : T \rightarrow X$ and $g : T \rightarrow Y$ are S -morphisms, we immediately have that

$$(5.3.1.2) \quad (f, g)_S = (f \times_S g) \circ \Delta_{T|S}.$$

The reader will note that the preceding definition and the results stated in (5.3.1) to (5.3.8) are true in any category, provided that the products used within exist in the category.

Proposition (5.3.2). — Let X and Y be S -preschemes; if we make the canonical identification between $(X \times Y) \times (X \times Y)$ and $(X \times X) \times (Y \times Y)$, then the morphism $\Delta_{X \times Y}$ is identified with $\Delta_X \times \Delta_Y$.

Proof. Indeed, if $p_1 : X \times X \rightarrow X$ and $q_1 : Y \times Y \rightarrow Y$ are the projections onto the first component, then the projection onto the first component $(X \times Y) \times (X \times Y) \rightarrow X \times Y$ is identified with $p_1 \times q_1$, and we have

$$(p_1 \times q_1) \circ (\Delta_X \times \Delta_Y) = (p_1 \circ \Delta_X) \times (q_1 \circ \Delta_Y) = 1_{X \times Y}$$

and we can argue similarly for the projections onto the second component. $\square$

Corollary (5.3.4). — For every base extension $S' \rightarrow S$, $\Delta_{X(S')}$ is canonically identified with $(\Delta_X)_{(S')}$.

Proof. It suffices to remark that $(X \times_S X)_{(S')}$ is canonically identified with $X_{(S')} \times_{S'} X_{(S')}$ (3.3.10). $\square$

Proposition (5.3.5). — Let X and Y be S -preschemes, and $\phi : S \rightarrow T$ a morphism of preschemes, which lets us consider every S -prescheme as a T -prescheme. Let $f : X \rightarrow S$ and $g : Y \rightarrow S$ ¹³ be the structure morphisms, p and q the projections of $X \times_S Y$, and $\pi = f \circ p = g \circ q$ the structure morphism $X \times_S Y \rightarrow S$. Then the diagram

$$(5.3.5.1) \quad \begin{array}{ccc} X \times_S Y & \xrightarrow{(p,q)_T} & X \times_T Y \\ \pi \downarrow & & \downarrow f \times_T g \\ S & \xrightarrow{\Delta_{S|T}} & S \times_T S \end{array}$$

commutes, and identifies $X \times_S Y$ with the product of the $(S \times_T S)$ -preschemes S and $X \times_T Y$, and the projections with π and $(p, q)_T$. $\square$

¹³[Trans.] The French source prints only $Y \rightarrow S$ here, omitting the name g ; the following identity $\pi = f \circ p = g \circ q$ identifies this second structure morphism as g .

Proof. By (3.4.3), we are led to proving the corresponding proposition *in the category of sets*, replacing X, Y , and S by $X(Z)_T, Y(Z)_T$, and $S(Z)_T$ (respectively), with Z being an arbitrary T -prescheme. But, in the category of sets, the proof is immediate and left to the reader. $\square$

Corollary (5.3.6). — *The morphism $(p, q)_T$ can be identified (letting $P = S \times_T S$) with $1_{X \times_T Y} \times_P \Delta_S$.*

Proof. This follows from (5.3.5) and (3.3.4). $\square$

Corollary (5.3.7). — *If $f : X \rightarrow Y$ is an S -morphism, then the diagram*

$$\begin{array}{ccc} X & \xrightarrow{(1_X, f)_S} & X \times_S Y \\ f \downarrow & & \downarrow f \times_S 1_Y \\ Y & \xrightarrow{\Delta_Y} & Y \times_S Y \end{array}$$

commutes, and identifies X with the product of the $(Y \times_S Y)$ -preschemes Y and $X \times_S Y$.

Proof. It suffices to apply (5.3.5), replacing S by Y , and T by S , and noting that $X \times_Y Y = X$ (3.3.3). $\square$

Proposition (5.3.8). — *For $f : X \rightarrow Y$ to be a monomorphism of preschemes, it is necessary and sufficient for $\Delta_{X|Y}$ to be an isomorphism from X to $X \times_Y X$.*

Proof. Indeed, f is a monomorphism if and only if, for every Y -prescheme Z , the corresponding map $f' : X(Z)_Y \rightarrow Y(Z)_Y$ is injective. Since $Y(Z)_Y$ consists of a single element, this is equivalent to saying that $X(Z)_Y$ has at most one element¹⁴, or, equivalently, that the diagonal map from $X(Z)_Y$ to $X(Z)_Y \times X(Z)_Y$ is a bijection. The latter product is exactly the set $(X \times_Y X)(Z)_Y$ (3.4.3.1), so this is equivalent to $\Delta_{X|Y}$ being an isomorphism. $\square$

Proposition (5.3.9). — *The diagonal morphism Δ_X is an immersion from X to $X \times_S X$.*

Proof. Indeed, since the continuous maps p_1 and Δ_X from the underlying spaces are such that $p_1 \circ \Delta_X$ is the identity, Δ_X is a homeomorphism from X to $\Delta_X(X)$. Similarly, the composite homomorphism $\mathcal{O}_X \rightarrow \mathcal{O}_{\Delta_X(X)} \rightarrow \mathcal{O}_X$ (composed of the homomorphisms corresponding to p_1 and Δ_X) is the identity, which means that the homomorphism corresponding to Δ_X is surjective; the proposition thus follows from (4.2.2). $\square$

We say that the subprescheme of $X \times_S X$ associated to the immersion Δ_X (4.2.1) is the *diagonal* of $X \times_S X$.

Corollary (5.3.10). — *Under the hypotheses of (5.3.5), $(p, q)_T$ is an immersion.*

Proof. This follows from (5.3.6) and (4.3.1). $\square$

We say (under the hypotheses of (5.3.5)) that $(p, q)_T$ is the *canonical immersion* of $X \times_S Y$ into $X \times_T Y$.

Corollary (5.3.11). — *Let X and Y be S -preschemes, and $f : X \rightarrow Y$ an S -morphism; then the graph morphism $\Gamma_f = (1_X, f)_S$ of f (3.3.14) is an immersion of X into $X \times_S Y$.*

Proof. This is a particular case of Corollary (5.3.10), where we replace S by Y , and T by S (cf. (5.3.7)). $\square$

The subprescheme of $X \times_S Y$ associated to the immersion Γ_f (4.2.1) is called the *graph* of the morphism f ; the subpreschemes of $X \times_S Y$ that are graphs of morphisms $X \rightarrow Y$ are characterised by the property that the restriction to such a subprescheme G of the projection $p_1 : X \times_S Y \rightarrow X$ is an isomorphism g from G to X : G is the graph of the morphism $p_2 \circ g^{-1}$, where p_2 is the projection $X \times_S Y \rightarrow Y$.

When we take, in particular, $X = S$, then the S -morphisms $S \rightarrow Y$ (which are exactly the S -sections of Y (2.5.5)) are equal to their graph morphisms; the subpreschemes of Y that are the graphs

¹⁴[Trans.] The French says that $X(Z)_Y$ is likewise reduced to one element; injectivity into the singleton $Y(Z)_Y$ yields only at most one element, allowing the empty case.

of S -sections (in other words, those that are isomorphic to S by the restriction of the structure morphism $Y \rightarrow S$) are then also called the *images* of these sections, or, by an abuse of language, the *S -sections* of Y .

Corollary (5.3.12). — *With the hypotheses and notation of (5.3.11), for every morphism $g : S' \rightarrow S$, let f' be the inverse image of f under g (3.3.7); then $\Gamma_{f'}$ is the inverse image of Γ_f under g .*

Proof. This is a particular case of (3.3.10.1). $\square$

Corollary (5.3.13). — *Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be morphisms; if $g \circ f$ is an immersion (resp. a local immersion), then so too is f .*

Proof. Indeed, f factors as $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$. Furthermore, p_2 can be identified with $(g \circ f) \times_Z 1_Y$ (3.3.4); if $g \circ f$ is an immersion (resp. a local immersion), then so too is p_2 ((4.3.1) and (4.5.5)), and since Γ_f is an immersion (5.3.11), we are done, by (4.2.4) (resp. (4.5.5)). $\square$

Corollary (5.3.14). — *Let $j : X \rightarrow Y$ and $g : X \rightarrow Z$ be S -morphisms. If j is an immersion (resp. a local immersion), then so too is $(j, g)_S$.*

Proof. Indeed, if $p : Y \times_S Z \rightarrow Y$ is the projection onto the first component, then we have $j = p \circ (j, g)_S$, and it suffices to apply (5.3.13). $\square$

Proposition (5.3.15). — *If $f : X \rightarrow Y$ is an S -morphism, then the diagram*

$$(5.3.15.1) \quad \begin{array}{ccc} X & \xrightarrow{\Delta_X} & X \times_S X \\ f \downarrow & & \downarrow f \times_S f \\ Y & \xrightarrow{\Delta_Y} & Y \times_S Y \end{array}$$

commutes (in other words, Δ_X is a functorial morphism in the category of preschemes).

Proof. The proof is immediate and left to the reader. $\square$

Corollary (5.3.16). — *If X is a subprescheme of Y , then the diagonal $\Delta_X(X)$ can be identified with a subprescheme of $\Delta_Y(Y)$, and the underlying space can be identified with*

$$\Delta_Y(Y) \cap p_1^{-1}(X) = \Delta_Y(Y) \cap p_2^{-1}(X)$$

(p_1 and p_2 being the projections of $Y \times_S Y$).

Proof. Applying (5.3.15) to the injection morphism $f : X \rightarrow Y$, we see that $f \times_S f$ is an immersion that identifies the underlying space of $X \times_S X$ with the subspace $p_1^{-1}(X) \cap p_2^{-1}(X)$ of $Y \times_S Y$ (4.3.1); further, if $z \in \Delta_Y(Y) \cap p_1^{-1}(X)$, then we have $z = \Delta_Y(y)$ and $y = p_1(z) \in X$, so $y = f(y)$, and $z = \Delta_Y(f(y))$ belongs to $\Delta_X(X)$ by the commutativity of (5.3.15.1). $\square$

Corollary (5.3.17). — *Let $f_1 : Y \rightarrow X$ and $f_2 : Y \rightarrow X$ be S -morphisms, and y a point of Y such that $f_1(y) = f_2(y) = x$, and such that the homomorphisms $k(x) \rightarrow k(y)$ corresponding to f_1 and f_2 are identical. Then, if $f = (f_1, f_2)_S$, the point $f(y)$ belongs to the diagonal $\Delta_{X|S}(X)$.*

Proof. The two homomorphisms $k(x) \rightarrow k(y)$ corresponding to f_i ($i = 1, 2$) define two S -morphisms $g_i : \text{Spec}(k(y)) \rightarrow \text{Spec}(k(x))$ such that the diagrams

$$\begin{array}{ccc} \text{Spec}(k(y)) & \xrightarrow{g_i} & \text{Spec}(k(x)) \\ \downarrow & & \downarrow \\ Y & \xrightarrow{f_i} & X \end{array}$$

commute. The diagram

$$\begin{array}{ccc} \text{Spec}(k(y)) & \xrightarrow{(g_1, g_2)_S} & \text{Spec}(k(x)) \times_S \text{Spec}(k(x)) \\ \downarrow & & \downarrow \\ Y & \xrightarrow{(f_1, f_2)_S} & X \times_S X \end{array}$$

thus also commutes. But it follows from the equality $g_1 = g_2$ that the image under $(g_1, g_2)_S$ of the unique point of $\text{Spec}(k(y))$ belongs to the diagonal of $\text{Spec}(k(x)) \times_S \text{Spec}(k(x))$; the conclusion then follows from (5.3.15). $\square$

5.4. Separated morphisms and separated preschemes

Definition (5.4.1). — We say that a morphism of preschemes $f : X \rightarrow Y$ is *separated* if the diagonal morphism $X \rightarrow X \times_Y X$ is a *closed immersion*; we then also say that X is a *separated prescheme* over Y , or a Y -*scheme*. We say that a prescheme X is separated if it is separated over $\text{Spec}(\mathbf{Z})$; we then also say that X is a *scheme*¹⁵ (cf. (5.5.7)).

By (5.3.9), for X to be separated over Y , it is necessary and sufficient for $\Delta_X(X)$ to be a *closed subspace* of the underlying space of $X \times_Y X$.

Proposition (5.4.2). — Let $S \rightarrow T$ be a separated morphism. If X and Y are S -preschemes, then the canonical immersion $X \times_S Y \rightarrow X \times_T Y$ (5.3.10) is closed.

Proof. Indeed, if we refer to the diagram in (5.3.5.1), we see that $(p, q)_T$ can be considered as being obtained from $\Delta_{S|T}$ by the base extension $f \times_T g : X \times_T Y \rightarrow S \times_T S$; the proposition then follows from (4.3.2). $\square$

Corollary (5.4.3). — Let Y be an S -scheme, and $f : X \rightarrow Y$ an S -morphism. Then the graph morphism $\Gamma_f : X \rightarrow X \times_S Y$ (5.3.11) is a closed immersion.

Proof. This is a particular case of (5.4.2), where we replace S by Y , and T by S . $\square$

Corollary (5.4.4). — Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be morphisms, with g separated. If $g \circ f$ is a closed immersion, then so too is f .

Proof. The proof using (5.4.3) is the same as that of (5.3.13) using (5.3.11). $\square$

I | 136

Corollary (5.4.5). — Let Z be an S -scheme, and $j : X \rightarrow Y$ and $g : X \rightarrow Z$ S -morphisms. If j is a closed immersion, then so too is $(j, g)_S : X \rightarrow Y \times_S Z$.

Proof. The proof using (5.4.4) is the same as that of (5.3.14) using (5.3.13). $\square$

Corollary (5.4.6). — If X is an S -scheme, then every S -section of X (2.5.5) is a closed immersion.

Proof. If $\phi : X \rightarrow S$ is the structure morphism, and $\psi : S \rightarrow X$ an S -section of X , it suffices to apply (5.4.5) to $\phi \circ \psi = 1_S$. $\square$

Corollary (5.4.7). — Let S be an integral prescheme with generic point s , and X an S -scheme. If two S -sections f and g are such that $f(s) = g(s)$, then $f = g$.

Proof. Indeed, if $x = f(s) = g(s)$, then the homomorphisms $k(x) \rightarrow k(s)$ corresponding to f and g are necessarily identical. If $h = (f, g)_S$, we thus deduce from (5.3.17) that $h(s)$ belongs to the diagonal $Z = \Delta_X(X)$; but since $S = \{s\}$, and since Z is closed by hypothesis, we have $h(S) \subset Z$. It then follows from (5.2.2) that h factors as $S \rightarrow Z \rightarrow X \times_S X$, and we thus conclude that $f = g$, by definition of the diagonal. $\square$

Remark (5.4.8). — If we suppose, conversely, that the conclusion of (5.4.3) is true when $f = 1_Y$, then we can conclude that Y is separated over S ; similarly, if we suppose that the conclusion of (5.4.5) applies to the two morphisms $Y \xrightarrow{\Delta_Y} Y \times_Z Y \xrightarrow{p_1} Y$, then we can conclude that Δ_Y is a closed immersion, and thus that Y is separated over Z ; finally, if we assume that the conclusion of (5.4.6) is true for the Y -section Δ_Y of the Y -prescheme $Y \times_S Y \rightarrow Y$, then this implies that Y is separated over S .

¹⁵[Trans.] We repeat here the warning given at the very start of this translation: the early versions of the EGA use *prescheme* to mean what is now usually called a *scheme*, and *scheme* for what is now usually called a *separated scheme*. Grothendieck himself later said that the more modern terminology was preferable, but we have decided to keep this translation ‘historically accurate’ by using the older nomenclature.

5.5. Separation criteria

Proposition (5.5.1). —

- (i) Every monomorphism of preschemes (and, in particular, every immersion) is a separated morphism.
- (ii) The composition of any two separated morphisms is separated.
- (iii) If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are separated S -morphisms, then $f \times_S g$ is separated.
- (iv) If $f : X \rightarrow Y$ is a separated S -morphism, then the S' -morphism $f_{(S')}$ is separated for every extension $S' \rightarrow S$ of the base prescheme.
- (v) If the composition $g \circ f$ is separated, then f is separated.
- (vi) For a morphism f to be separated, it is necessary and sufficient for f_{red} (5.1.5) to be separated.

Proof. Note that (i) is an immediate consequence of (5.3.8). If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are morphisms, then the diagram

$$(5.5.1.1) \quad \begin{array}{ccc} X & \xrightarrow{\Delta_{X|Z}} & X \times_Z X \\ & \searrow \Delta_{X|Y} & \nearrow j \\ & X \times_Y X & \end{array}$$

commutes (where j denotes the canonical immersion (5.3.10)), as can be immediately verified. If f and g are separated, then $\Delta_{X|Y}$ is a closed immersion by definition, and j is a closed immersion by (5.4.2), whence $\Delta_{X|Z}$ is a closed immersion by (4.2.4), which proves (ii). Given (i) and (ii), (iii) and (iv) are equivalent (3.5.1), so it suffices to prove (iv). But $X_{(S')} \times_{Y_{(S')}} X_{(S')}$ is canonically identified with $(X \times_Y X) \times_{Y_{(S')}} Y_{(S')}$ by (3.3.11) and (3.3.9.1), and we immediately see that the diagonal morphism $\Delta_{X_{(S')}}$ can then be identified with $\Delta_X \times_Y 1_{Y_{(S')}}$; the proposition then follows from (4.3.1).

I | 137

To prove (v), consider, as in (5.3.13), the factorisation $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$ of f , noting that $p_2 = (g \circ f) \times_Z 1_Y$; the hypothesis (that $g \circ f$ is separated) implies that p_2 is separated, by (iii) and (i), and, since Γ_f is an immersion, Γ_f is separated, by (i), whence f is separated, by (ii). Finally, to prove (vi), recall that the preschemes $X_{\text{red}} \times_{Y_{\text{red}}} X_{\text{red}}$ and $X_{\text{red}} \times_Y X_{\text{red}}$ are canonically identified with one another (5.1.7); if we denote by j the injection $X_{\text{red}} \rightarrow X$, then the diagram

$$\begin{array}{ccc} X_{\text{red}} & \xrightarrow{\Delta_{X_{\text{red}}}} & X_{\text{red}} \times_Y X_{\text{red}} \\ j \downarrow & & \downarrow j \times_Y j \\ X & \xrightarrow{\Delta_X} & X \times_Y X \end{array}$$

commutes (5.3.15), and the proposition follows from the fact that the vertical arrows are homeomorphisms of the underlying spaces (4.3.1). $\square$

Corollary (5.5.2). — If $f : X \rightarrow Y$ is separated, then the restriction of f to any subprescheme of X is separated.

Proof. This follows from (5.5.1, i and ii). $\square$

Corollary (5.5.3). — If X and Y are S -preschemes such that Y is separated over S , then $X \times_S Y$ is separated over X .

Proof. This is a particular case of (5.5.1, iv). $\square$

Proposition (5.5.4). — Let X be a prescheme, and assume that its underlying space is a finite union of closed subsets X_k ($1 \leq k \leq n$); for each k , consider the reduced subprescheme of X that has X_k as its underlying space (5.2.1), and denote this again by X_k . Let $f : X \rightarrow Y$ be a morphism, and, for each k , let Y_k be a closed subset of Y such that $f(X_k) \subset Y_k$; we again denote by Y_k the reduced subprescheme of Y that has Y_k as its underlying space, so that the restriction $X_k \rightarrow Y$ of f to X_k factors as $X_k \xrightarrow{f_k} Y_k \rightarrow Y$ (5.2.2). For f to be separated, it is necessary and sufficient for all the f_k to be separated.

Proof. The necessity follows from (5.5.1, i, ii, and v). Conversely, if the condition of the statement is satisfied, then each of the restrictions $X_k \rightarrow Y$ of f is separated (5.5.1, (i) and (ii)); if p_1 and p_2 are the projections of $X \times_Y X$, then the subspace $\Delta_{X_k}(X_k)$ can be identified with the subspace $\Delta_X(X) \cap p_1^{-1}(X_k)$ of the underlying space of $X \times_Y X$ (5.3.16); these subspaces are closed in $X \times_Y X$, and thus so too is their union $\Delta_X(X)$. $\square$

Suppose, in particular, that the X_k are the *irreducible components* of X ; then we can suppose that the Y_k are the irreducible components of Y (0, 2.1.5); Proposition (5.5.4) therefore reduces, in this case, the notion of separation to the case of *integral* preschemes (2.1.7).

I | 138

Proposition (5.5.5). — *Let (Y_λ) be an open cover of a prescheme Y ; for a morphism $f : X \rightarrow Y$ to be separated, it is necessary and sufficient for each of its restrictions $f^{-1}(Y_\lambda) \rightarrow Y_\lambda$ to be separated.*

Proof. If we set $X_\lambda = f^{-1}(Y_\lambda)$, everything reduces, by taking (4.2.4, b) and the identification of the products $X_\lambda \times_Y X_\lambda$ and $X_\lambda \times_{Y_\lambda} X_\lambda$ (3.2.5) into account, to proving that the $X_\lambda \times_Y X_\lambda$ form a cover of $X \times_Y X$. But if we set $Y_{\lambda\mu} = Y_\lambda \cap Y_\mu$ and $X_{\lambda\mu} = X_\lambda \cap X_\mu = f^{-1}(Y_{\lambda\mu})$, then $X_\lambda \times_Y X_\mu$ can be identified with the product $X_{\lambda\mu} \times_{Y_{\lambda\mu}} X_{\lambda\mu}$ (3.2.6.4), and so also with $X_{\lambda\mu} \times_Y X_{\lambda\mu}$ (3.2.5), and finally with an open subset of $X_\lambda \times_Y X_\lambda$, which proves our claim (3.2.7). $\square$

Proposition (5.5.4) allows us, by taking an affine open cover of Y , to reduce the study of separated morphisms to that of separated morphisms taking values in affine schemes.

Proposition (5.5.6). — *Let Y be an affine scheme, X a prescheme, and (U_α) a cover of X by affine open subsets. For a morphism $f : X \rightarrow Y$ to be separated, it is necessary and sufficient for $U_\alpha \cap U_\beta$ to be, for every pair of indices (α, β) , an affine open subset, and for the ring $\Gamma(U_\alpha \cap U_\beta, \mathcal{O}_X)$ to be generated by the union of the canonical images of the rings $\Gamma(U_\alpha, \mathcal{O}_X)$ and $\Gamma(U_\beta, \mathcal{O}_X)$.*

Proof. The $U_\alpha \times_Y U_\beta$ form an open cover of $X \times_Y X$ (3.2.7); denoting the projections of $X \times_Y X$ by p and q , we have

$$\begin{aligned} \Delta_X^{-1}(U_\alpha \times_Y U_\beta) &= \Delta_X^{-1}(p^{-1}(U_\alpha) \cap q^{-1}(U_\beta)) \\ &= \Delta_X^{-1}(p^{-1}(U_\alpha)) \cap \Delta_X^{-1}(q^{-1}(U_\beta)) = U_\alpha \cap U_\beta; \end{aligned}$$

so everything reduces to proving that the restriction of Δ_X to $U_\alpha \cap U_\beta$ is a closed immersion into $U_\alpha \times_Y U_\beta$. But this restriction is exactly $(j_\alpha, j_\beta)_Y$, where j_α (resp. j_β) denotes the injection morphism from $U_\alpha \cap U_\beta$ to U_α (resp. U_β), as follows from the definitions. Since $U_\alpha \times_Y U_\beta$ is an affine scheme whose ring is canonically isomorphic to $\Gamma(U_\alpha, \mathcal{O}_X) \otimes_{\Gamma(Y, \mathcal{O}_Y)} \Gamma(U_\beta, \mathcal{O}_X)$ (3.2.2), we see that $U_\alpha \cap U_\beta$ must be an affine scheme, and that the map $h_\alpha \otimes h_\beta \mapsto h_\alpha h_\beta$ from the ring $A(U_\alpha \times_Y U_\beta)$ to $\Gamma(U_\alpha \cap U_\beta, \mathcal{O}_X)$ must be surjective (4.2.3), which finishes the proof. $\square$

Corollary (5.5.7). — *An affine scheme is separated (and is thus a scheme, which justifies the terminology of (5.4.1)).*

Corollary (5.5.8). — *Let Y be an affine scheme; for $f : X \rightarrow Y$ to be a separated morphism, it is necessary and sufficient for X to be separated (in other words, for X to be a scheme).*

Proof. Indeed, we see that the criterion of (5.5.6) does not depend on f . $\square$

Corollary (5.5.9). — *For a morphism $f : X \rightarrow Y$ to be separated, it is necessary for the induced prescheme $f^{-1}(U)$ to be separated for every open subset U on which Y induces a separated prescheme, and it is sufficient for this to hold for every affine open subset $U \subset Y$.*

Proof. The necessity of the condition follows from (5.5.4) and (5.5.1, ii); the sufficiency follows from (5.5.4) and (5.5.8), taking into account the existence of affine open covers of Y . $\square$

In particular, if X and Y are affine schemes, then *every* morphism $X \rightarrow Y$ is separated.

Proposition (5.5.10). — *Let Y be a scheme, and $f : X \rightarrow Y$ a morphism. For every affine open subset U of X , and every affine open subset V of Y , $U \cap f^{-1}(V)$ is affine.*

Proof. Let p_1 and p_2 be the projections of $X \times_Z Y$; the subspace $U \cap f^{-1}(V)$ is the image of $\Gamma_f(X) \cap p_1^{-1}(U) \cap p_2^{-1}(V)$ under p_1 . But $p_1^{-1}(U) \cap p_2^{-1}(V)$ can be identified with the underlying space of the $\square$